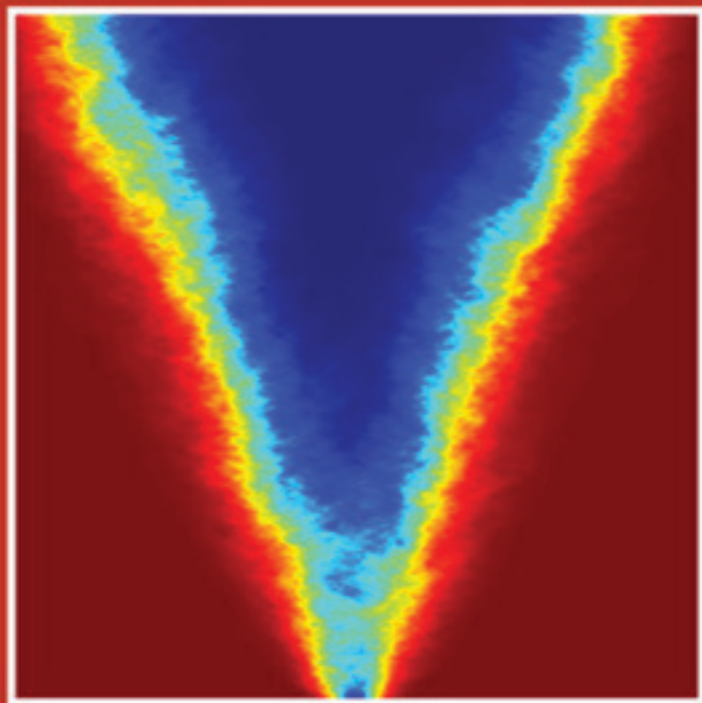


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An Introduction to Computational Stochastic PDEs



GABRIEL J. LORD
CATHERINE E. POWELL
TONY SHARDLOW

AN INTRODUCTION TO COMPUTATIONAL STOCHASTIC PDES

This book gives a comprehensive introduction to numerical methods and analysis of stochastic processes, random fields and stochastic differential equations, and offers graduate students and researchers powerful tools for understanding uncertainty quantification for risk analysis. Coverage includes traditional stochastic ordinary differential equations with white noise forcing, strong and weak approximation and the multilevel Monte Carlo method. Later chapters apply the theory of random fields to the numerical solution of elliptic PDEs with correlated random data, discuss the Monte Carlo method and introduce stochastic Galerkin finite element methods. Finally, stochastic parabolic PDEs are developed.

Assuming little previous exposure to probability and statistics, theory is developed in tandem with state-of-the-art computational methods through worked examples, exercises, theorems and proofs. The set of MATLAB codes included (and downloadable) allows readers to perform computations themselves and solve the test problems discussed. Practical examples are drawn from finance, mathematical biology, neuroscience, fluid flow modelling and materials science.

GABRIEL J. LORD is a professor in the Maxwell Institute, Department of Mathematics, at Heriot-Watt University. He has worked on stochastic PDEs and applications for more than ten years. He is the coeditor of *Stochastic Methods in Neuroscience* (with C. Laing) and has organised a number of international meetings in the field. He has served as an Associate Editor for the *SIAM Journal on Scientific Computing* and the *SIAM/ASA Journal on Uncertainty Quantification*.

CATHERINE E. POWELL is a senior lecturer in applied mathematics and numerical analysis at the University of Manchester. She has worked in the field of stochastic PDEs and uncertainty quantification for more than ten years. Together with Tony Shardlow, she initiated the NASPDE (Numerical Analysis of Stochastic PDEs) series of meetings. She has also served as an Associate Editor for the *SIAM/ASA Journal on Uncertainty Quantification*.

TONY SHARDLOW has been working in the numerical analysis group at the University of Bath since 2012. Before that, he held appointments at the universities of Manchester, Durham, Oxford and Minnesota. He completed his PhD in scientific computing and computational mathematics at Stanford University in 1997.

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GABRIEL J. LORD

Heriot-Watt University, Edinburgh

CATHERINE E. POWELL

University of Manchester

TONY SHARDLOW

University of Bath



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32 Avenue of the Americas, New York, NY 10013-2473, USA

Cambridge University Press is part of the University of Cambridge.

It furthers the University's mission by disseminating knowledge in the pursuit of education, learning and research at the highest international levels of excellence.

www.cambridge.org

Information on this title: www.cambridge.org/9780521728522

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First published 2014

Printed in the United States of America

A catalog record for this publication is available from the British Library.

Library of Congress Cataloging in Publication Data

Lord, Gabriel J., author.

An introduction to computational stochastic PDEs / Gabriel J. Lord, Heriot-Watt University, Edinburgh, Catherine E. Powell, University of Manchester, Tony Shardlow, University of Bath.

pages cm – (Cambridge texts in applied mathematics; 50)

Includes bibliographical references and index.

ISBN 978-0-521-89990-1 (hardback) – ISBN 978-0-521-72852-2 (paperback)

1. Stochastic partial differential equations. I. Powell, Catherine E., author.

II. Shardlow, Tony, author. III. Title.

QA274.25.L67 2014

519.2'2–dc23 2014005535

ISBN 978-0-521-89990-1 Hardback

ISBN 978-0-521-72852-2 Paperback

Additional resources for this publication at www.cambridge.org/9780521728522

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Preface

Techniques for solving many of the differential equations traditionally used by applied mathematicians to model phenomena such as fluid flow, neural dynamics, electromagnetic scattering, tumour growth, telecommunications, phase transitions, etc. are now mature. Parameters within those models (e.g., material properties, boundary conditions, forcing terms, domain geometries) are often assumed to be known exactly, even when it is clear that is not the case. In the past, mathematicians were unable to incorporate noise and/or uncertainty into models because they were constrained both by the lack of computational resources and the lack of research into stochastic analysis. These are no longer good excuses. The rapid increase in computing power witnessed in recent decades allows the extra level of complexity induced by uncertainty to be incorporated into numerical simulations. Moreover, there are a growing number of researchers working on stochastic partial differential equations (PDEs) and their results are continually improving our theoretical understanding of the behaviour of stochastic systems. The transition from working with purely deterministic systems to working with stochastic systems is understandably daunting for recent graduates who have majored in applied mathematics. It is perhaps even more so for established researchers who have not received any training in probability theory and stochastic processes. We hope this book bridges this gap and will provide training for a new generation of researchers — that is, you.

This text provides a friendly introduction and practical route into the numerical solution and analysis of stochastic PDEs. It is suitable for mathematically grounded graduates who wish to learn about stochastic PDEs and numerical solution methods. The book will also serve established researchers who wish to incorporate uncertainty into their mathematical models and seek an introduction to the latest numerical techniques. We assume knowledge of undergraduate-level mathematics, including some basic analysis and linear algebra, but provide background material on probability theory and numerical methods for solving differential equations. Our treatment of model problems includes analysis, appropriate numerical methods and a discussion of practical implementation. `MATLAB` is a convenient computer environment for numerical scientific computing and is used throughout the book to solve examples that illustrate key concepts. We provide code to implement the algorithms on model problems, and sample code is available from the authors' or the publisher's website*. Each chapter concludes with exercises, to help the reader study and become more

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familiar with the concepts involved, and a section of notes, which contains pointers and references to the latest research directions and results.

The book is divided into three parts, as follows.

Part One: Deterministic Differential Equations We start with a deterministic or non-random outlook and introduce preliminary background material on functional analysis, numerical analysis, and differential equations. [Chapter 1](#) reviews linear analysis and introduces Banach and Hilbert spaces, as well as the Fourier transform and other key tools from Fourier analysis. [Chapter 2](#) treats elliptic PDEs, starting with a two-point boundary-value problem (BVP), and develops Galerkin approximation and the finite element method. [Chapter 3](#) develops numerical methods for initial-value problems for ordinary differential equations (ODEs) and a class of semilinear PDEs that includes reaction–diffusion equations. We develop finite difference methods and spectral and finite element Galerkin methods. [Chapters 2 and 3](#) include not only error analysis for selected numerical methods but also MATLAB implementations for test problems that illustrate numerically the theoretical orders of convergence.

Part Two: Stochastic Processes and Random Fields Here we turn to probability theory and develop the theory of stochastic processes (one parameter families of random variables) and random fields (multi-parameter families of random variables). Stochastic processes and random fields are used to model the uncertain inputs to the differential equations studied in Part Three and are also the appropriate way to interpret the corresponding solutions. [Chapter 4](#) starts with elementary probability theory, including random variables, limit theorems, and sampling methods. The Monte Carlo method is introduced and applied to a differential equation with random initial data. [Chapters 5–7](#) then develop theory and computational methods for stochastic processes and random fields. Specific stochastic processes discussed include Brownian motion, white noise, the Brownian bridge, and fractional Brownian motion. In [Chapters 6 and 7](#), we pay particular attention to the important special classes of stationary processes and isotropic random fields. Simulation methods are developed, including a quadrature scheme, the turning bands method, and the highly efficient FFT-based circulant embedding method. The theory of these numerical methods is developed alongside practical implementations in MATLAB.

Part Three: Stochastic Differential Equations There are many ways to incorporate stochastic effects into differential equations. In the last part of the book, we consider three classes of stochastic model problems, each of which can be viewed as an extension to a deterministic model introduced in Part One. These are:

Chapter 8	ODE (3.6)	+ white noise forcing
Chapter 9	Elliptic BVP (2.1)	+ correlated random data
Chapter 10	Semilinear PDE (3.39)	+ space–time noise forcing

Note the progression from models for time t and sample variable ω in [Chapter 8](#), to models for space $\mathbf{x}$ and ω in [Chapter 9](#), and finally to models for $t, \mathbf{x}, \omega$ in [Chapter 10](#). In each case, we adapt the techniques from [Chapters 2 and 3](#) to show that the problems are well posed and to develop numerical approximation schemes. MATLAB implementations are also discussed. Brownian motion is key to developing the time-dependent problems with white noise forcing considered in [Chapters 8 and 10](#) using the Itô calculus. It is these types

of differential equations that are traditionally known as stochastic differential equations (SODEs and SPDEs). In [Chapter 9](#), however, we consider elliptic BVPs with both a forcing term and coefficients that are represented by random fields *not* of white noise type. Many authors prefer to reserve the term ‘stochastic PDE’ only for PDEs forced by white noise. We interpret it more broadly, however, and the title of this book is intended to incorporate PDEs with data and/or forcing terms described by both white noise (which is uncorrelated) and correlated random fields. The analytical tools required to solve these two types of problems are, of course, very different and we give an overview of the key results.

[Chapter 8](#) introduces the class of stochastic ordinary differential equations (SODEs) consisting of ODEs with white noise forcing, discusses existence and uniqueness of solutions in the sense of Itô calculus, and develops the Euler–Maruyama and Milstein approximation schemes. Strong approximation (of samples of the solution) and weak approximation (of averages) are discussed, as well as the multilevel Monte Carlo method. [Chapter 9](#) treats elliptic BVPs with correlated random data on two-dimensional spatial domains. These typically arise in the modelling of fluid flow in porous media. Solutions are also correlated random fields and, here, do not depend on time. To begin, we consider log-normal coefficients. After sampling the input data, we study weak solutions to the resulting deterministic problems and apply the Galerkin finite element method. The Monte Carlo method is then used to estimate the mean and variance. By approximating the data using Karhunen–Loève expansions, the stochastic PDE problem may also be converted to a deterministic one on a (possibly) high-dimensional parameter space. After setting up an appropriate weak formulation, the stochastic Galerkin finite element method (SGFEM), which couples finite elements in physical space with global polynomial approximation on a parameter space, is developed in detail. [Chapter 10](#) develops stochastic parabolic PDEs, such as reaction–diffusion equations forced by a space–time Wiener process, and we discuss (strong) numerical approximation in space and in time. Model problems arising in the fields of neuroscience and fluid dynamics are included.

The number of questions that can be asked of stochastic PDEs is large. Broadly speaking, they fall into two categories: *forward problems* (sampling the solution, determining exit times, computing moments, etc.) and *inverse problems* (e.g., fitting a model to a set of observations). In this book, we focus on forward problems for specific model problems. We pay particular attention to elliptic PDEs with coefficients given by correlated random fields and reaction–diffusion equations with white noise forcing. We also focus on methods to compute individual samples of solutions and to compute moments (means, variances) of functionals of the solutions. Many other stochastic PDE models are neglected (hyperbolic problems, random domains, forcing by Levy processes, to name a few) as are many important questions (exit time problems, long-time simulation, filtering). However, this book covers a wide range of topics necessary for studying these problems and will leave the reader well prepared to tackle the latest research on the numerical solution of a wide range of stochastic PDEs.

1

Linear Analysis

This chapter introduces theoretical tools for studying stochastic differential equations in later chapters. §1.1 and §1.2 review Banach and Hilbert spaces, the mathematical structures given to sets of random variables and the natural home for solutions of differential equations. §1.3 reviews the theory of linear operators, especially the spectral theory of compact and symmetric operators, and §1.4 reviews Fourier analysis.

1.1 Banach spaces C^r and L^p

Banach and Hilbert spaces are fundamental to the analysis of differential equations and random processes. This section treats Banach spaces, reviewing first the notions of norm, convergence, and completeness before giving [Definition 1.7](#) of a Banach space. We assume readers are familiar with real and complex vector spaces.

Definition 1.1 (norm) A norm $\|\cdot\|$ is a function from a real (respectively, complex) vector space X to $\mathbb{R}^+$ such that

- (i) $\|u\| = 0$ if and only if $u = 0$,
- (ii) $\|\lambda u\| = |\lambda| \|u\|$ for all $u \in X$ and $\lambda \in \mathbb{R}$ (resp., $\mathbb{C}$), and
- (iii) $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in X$ (triangle inequality).

A *normed vector space* $(X, \|\cdot\|)$ is a vector space X with a norm $\|\cdot\|$. If only conditions (ii) and (iii) hold, $\|\cdot\|$ is called a *semi-norm* and denoted $|\cdot|_X$.

Example 1.2 $(\mathbb{R}^d, \|\cdot\|_2)$ is a normed vector space with

$$\|u\|_2 := \left(|u_1|^2 + \cdots + |u_d|^2\right)^{1/2}$$

for the column vector $u = [u_1, \dots, u_d]^T \in \mathbb{R}^d$, where $|\cdot|$ denotes absolute value. More generally, $\|u\|_\infty := \max\{|u_1|, \dots, |u_d|\}$ and $\|u\|_p := (|u_1|^p + \cdots + |u_d|^p)^{1/p}$ for $p \geq 1$ is a norm and $(\mathbb{R}^d, \|\cdot\|_p)$ is a normed vector space. When $d = 1$, these norms are all equal to the absolute value.

Definition 1.3 (domain) A domain D is a non-trivial, connected, open subset of $\mathbb{R}^d$ and a domain is bounded if $D \subset \{x \in \mathbb{R}^d : \|x\|_2 \leq R\}$ for some $R > 0$. The boundary of a domain is denoted ∂D and we always assume the boundary is piecewise smooth (e.g., the boundary of a polygon or a sphere).

Example 1.4 (continuous functions) For a subset $D \subset \mathbb{R}^d$, let $C(D)$ denote the set of real-valued continuous functions on D . If D is a domain, functions in $C(D)$ may be unbounded. However, functions in $C(\bar{D})$, where $\bar{D}$ is the closure of D , are bounded and $(C(\bar{D}), \|\cdot\|_\infty)$ is a normed vector space with the supremum norm,

$$\|u\|_\infty := \sup_{x \in \bar{D}} |u(x)|, \quad u \in C(\bar{D}).$$

A norm $\|\cdot\|$ on a vector space X measures the size of elements in X and provides a notion of convergence: for $u, u_n \in X$, we write $u = \lim_{n \rightarrow \infty} u_n$ or $u_n \rightarrow u$ as $n \rightarrow \infty$ in $(X, \|\cdot\|)$ if $\|u_n - u\| \rightarrow 0$ as $n \rightarrow \infty$. For example, the notion of convergence on $C(\bar{D})$ is known as uniform convergence.

Definition 1.5 (uniform and pointwise convergence) We say $u_n \in C(\bar{D})$ converges *uniformly* to a limit u if $\|u_n - u\|_\infty \rightarrow 0$ as $n \rightarrow \infty$. Explicitly, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that, for all $x \in \bar{D}$ and all $n \geq N$, $|u_n(x) - u(x)| < \epsilon$. In uniform convergence, N depends only on ϵ . This should be contrasted with the notion of pointwise convergence, which applies to all functions $u_n : D \rightarrow \mathbb{R}$. We say $u_n \rightarrow u$ *pointwise* if, for every $x \in D$ and every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|u_n(x) - u(x)| < \epsilon$. In pointwise convergence, N may depend both on ϵ and x .

There are many techniques, both computational and analytical, for finding approximate solutions $u_n \in X$ to mathematical problems posed on a vector space X . When u_n is a Cauchy sequence and X is complete, u_n converges to some $u \in X$, the so-called limit point, and this is often key in showing a mathematical model is well posed and proving the existence of a solution.

Definition 1.6 (Cauchy sequence, complete) Consider a normed vector space $(X, \|\cdot\|)$. A sequence $u_n \in X$ for $n \in \mathbb{N}$ is called a *Cauchy sequence* if, for all $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\|u_n - u_m\| < \epsilon \quad \text{for all } n, m \geq N.$$

A normed vector space $(X, \|\cdot\|)$ is said to be *complete* if every Cauchy sequence u_n in X converges to a limit point $u \in X$. In other words, there exists a $u \in X$ such that $\|u_n - u\| \rightarrow 0$ as $n \rightarrow \infty$.

Definition 1.7 (Banach space) A *Banach space* is a complete normed vector space.

Example 1.8 $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^d, \|\cdot\|_p)$ for $1 \leq p \leq \infty$ are Banach spaces.

Example 1.9 $(C(\bar{D}), \|\cdot\|_\infty)$ is a Banach space if D is bounded. See [Exercise 1.1](#). If D is unbounded, the set of bounded continuous functions $C_b(D)$ on D gives a Banach space.

The contraction mapping theorem is used in [Chapters 3, 8, and 10](#) to prove the existence and uniqueness of solutions to initial-value problems.

Theorem 1.10 (contraction mapping) Let Y be a non-empty closed subset of the Banach space $(X, \|\cdot\|)$. Consider a mapping $\mathcal{J} : Y \rightarrow Y$ such that, for some $\mu \in (0, 1)$,

$$\|\mathcal{J}u - \mathcal{J}v\| \leq \mu \|u - v\|, \quad \text{for all } u, v \in Y. \quad (1.1)$$

There exists a unique fixed point of $\mathcal{J}$ in Y ; that is, there is a unique $u \in Y$ such that $\mathcal{J}u = u$.

Proof Fix $u_0 \in Y$ and consider $u_n = \mathcal{J}^n u_0$ (the n th iterate of u_0 under application of $\mathcal{J}$). The sequence u_n is easily shown to be Cauchy in Y using (1.1) and therefore converges to a limit $u \in Y$ because Y is complete (as a closed subset of X). Now $u_n \rightarrow u$ and hence $u_{n+1} = \mathcal{J}u_n \rightarrow \mathcal{J}u$ as $n \rightarrow \infty$. We conclude that u_n converges to a fixed point of $\mathcal{J}$.

If $u, v \in Y$ are both fixed points of $\mathcal{J}$, then $\mathcal{J}u - \mathcal{J}v = u - v$. But (1.1) holds and hence $u = v$ and the fixed point is unique. $\square$

Spaces of continuously differentiable functions

The smoothness, also called regularity, of a function is described by its derivatives and we now define spaces of functions with a given number of continuous derivatives. For a domain $D \subset \mathbb{R}^d$ and Banach space $(Y, \|\cdot\|_Y)$, consider a function $u: D \rightarrow Y$. We denote the partial derivative operator with respect to x_j by $\mathcal{D}_j := \frac{\partial}{\partial x_j}$. Given a multi-index $\alpha = (\alpha_1, \dots, \alpha_d)$, we define $|\alpha| := \alpha_1 + \dots + \alpha_d$ and $\mathcal{D}^\alpha := \mathcal{D}_1^{\alpha_1} \dots \mathcal{D}_d^{\alpha_d}$, so that

$$\mathcal{D}^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

Definition 1.11 (continuous functions)

- (i) $C(D, Y)$ is the set of continuous functions $u: D \rightarrow Y$. If D is bounded, we equip $C(\bar{D}, Y)$ with the norm

$$\|u\|_\infty := \sup_{\mathbf{x} \in \bar{D}} \|u(\mathbf{x})\|_Y. \quad (1.2)$$

- (ii) $C^r(D, Y)$ with $r \in \mathbb{N}$ is the set of functions $u: D \rightarrow Y$ such that $\mathcal{D}^\alpha u \in C(D, Y)$ for $|\alpha| \leq r$; that is, functions whose derivatives up to and including order r are continuous. We equip $C^r(\bar{D}, Y)$ with the norm

$$\|u\|_{C^r(\bar{D}, Y)} := \sum_{0 \leq |\alpha| \leq r} \|\mathcal{D}^\alpha u\|_\infty.$$

We abbreviate the notation so that $C(D, \mathbb{R})$ is denoted by $C(D)$ and $C^r(D, \mathbb{R})$ by $C^r(D)$.

Proposition 1.12 *If the domain D is bounded, $C(\bar{D}, Y)$ and $C^r(\bar{D}, Y)$ are Banach spaces.*

Proof The case of $C(\bar{D})$ is considered in [Exercise 1.1](#). $\square$

The following sets of continuous functions, which are not provided with a norm, are also useful.

Definition 1.13 (infinitely differentiable functions)

- (i) $C^\infty(D, Y)$ is the set $\bigcap_{r \in \mathbb{N}} C^r(D, Y)$ of infinitely differentiable functions from D to Y .
(ii) $C_c^\infty(D, Y)$ is the set of $u \in C^\infty(D, Y)$ such that $\text{supp } u$ is a compact subset of D , where the support $\text{supp } u$ denotes the closure of $\{\mathbf{x} \in D: u(\mathbf{x}) \neq 0\}$. (The definition of compact is recalled in [Definition 1.66](#)).

The spaces $C^r(D, Y)$ specify the regularity of a function via the number, r , of continuous derivatives. More refined concepts of regularity include Hölder and Lipschitz regularity.

Definition 1.14 (Hölder and Lipschitz) Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces. A function $u: X \rightarrow Y$ is *Hölder continuous* with constant $\gamma \in (0, 1]$ if there is a constant $L > 0$ so that

$$\|u(x_1) - u(x_2)\|_Y \leq L \|x_1 - x_2\|_X^\gamma, \quad \forall x_1, x_2 \in X.$$

If the above holds with $\gamma = 1$, then u is *Lipschitz continuous* or *globally Lipschitz continuous* to stress that L is uniform for $x_1, x_2 \in X$. The space $C^{r,\gamma}(D)$ is the set of functions in $C^r(D)$ whose r th derivatives are Hölder continuous with exponent γ .

Lebesgue integrals and measurability

The Lebesgue integral is an important generalisation of the Riemann integral. For a function $u: [a, b] \rightarrow \mathbb{R}$, the Riemann integral $\int_a^b u(x) dx$ is given by a limit of sums $\sum_{j=0}^{N-1} u(\xi_j)(x_{j+1} - x_j)$ for points $\xi_j \in [x_j, x_{j+1}]$, with respect to refinement of the partition $a = x_0 < \dots < x_N = b$. In other words, u is approximated by a piecewise constant function, whose integral is easy to evaluate, and a limiting process defines the integral of u . The Lebesgue integral is also defined by a limit, but instead of piecewise constant approximations on a partition of $[a, b]$, approximations constant on *measurable sets* are used.

Let 1_F be the indicator function of a set F so

$$1_F(x) := \begin{cases} 1, & x \in F, \\ 0, & x \notin F. \end{cases}$$

Suppose that $\{F_j\}$ are measurable sets in $[a, b]$ (see [Definition 1.15](#)) and $\mu(F_j)$ denotes the measure of F_j (e.g., if $F_j = [a, b]$ then $\mu([a, b]) = |b - a|$). The Lebesgue integral of u with respect to the measure μ is defined via

$$\int_a^b u(x) d\mu(x) = \lim \sum_j u_j \mu(F_j),$$

where the limit is taken as the functions $\sum_j u_j 1_{F_j}(x)$ converge to $u(x)$. The idea is illustrated in [Figure 1.1](#), where the function $u(x)$ is approximated by $\sum_{i=1}^3 u_i 1_{F_i}(x)$ for

$$\begin{aligned} F_1 &= u^{-1}([-1, -0.5]), & F_2 &= u^{-1}((-0.5, 0.5]), & F_3 &= u^{-1}((0.5, 1]), \\ u_1 &= -0.8, & u_2 &= 0, & u_3 &= 0.8. \end{aligned}$$

Here, $u^{-1}([a, b]) := \{x \in \mathbb{R} : u(x) \in [a, b]\}$. To precisely define the Lebesgue integral, we must first form a collection of subsets $\mathcal{F}$ that we can measure.

Definition 1.15 (σ -algebra) A set $\mathcal{F}$ of subsets of a set X is a σ -algebra if

- (i) the empty set $\{\} \in \mathcal{F}$,
- (ii) the complement $F^c := \{x \in X : x \notin F\} \in \mathcal{F}$ for all $F \in \mathcal{F}$, and
- (iii) the union $\cup_{j \in \mathbb{N}} F_j \in \mathcal{F}$ for $F_j \in \mathcal{F}$.

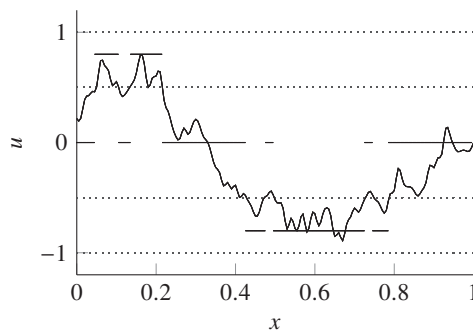


Figure 1.1 We approximate a function $u(x)$ by the simple function $\sum_{j=1}^3 u_j 1_{F_j}(x)$. The sets F_j arise by partitioning the y -axis and using the pullback sets, which are measurable when u is measurable. In contrast to the Riemann integral, F_j are not simple intervals.

Thus, a σ -algebra is a collection of subsets of X that contains the empty set and is closed under forming complements and countable unions. Any $F \in \mathcal{F}$ is known as a *measurable set* and the pair $(X, \mathcal{F})$ is known as a *measurable space*.

It is natural to ask why the power set of X (i.e., the set of all subsets of X) is not chosen for $\mathcal{F}$. It turns out, for example using Vitali sets or in the Banach–Tarski paradox, that non-intuitive effects arise through measuring every set.

For topological spaces like $\mathbb{R}^d$ or Banach spaces, $\mathcal{F}$ is chosen to be the smallest σ -algebra that contains all open sets; this is known as the *Borel σ -algebra*.

Definition 1.16 (Borel σ -algebra) For a topological space Y , $\mathcal{B}(Y)$ denotes the Borel σ -algebra and equals the smallest σ -algebra containing all open subsets of Y .

Definition 1.17 (measure) A measure μ on a measurable space $(X, \mathcal{F})$ is a mapping from $\mathcal{F}$ to $\mathbb{R}^+ \cup \{\infty\}$ such that

- (i) the empty set has measure zero, $\mu(\{\}) = 0$, and
- (ii) $\mu(\cup_{j \in \mathbb{N}} F_j) = \sum_{j \in \mathbb{N}} \mu(F_j)$ if $F_j \in \mathcal{F}$ are disjoint (i.e., $F_k \cap F_j = \{\}$ for $k \neq j$).

Together $(X, \mathcal{F}, \mu)$ form a *measure space*. We say the measure space is *σ -finite* if $X = \cup_{j=1}^{\infty} F_j$ for some $F_j \in \mathcal{F}$ with $\mu(F_j) < \infty$. A set $F \in \mathcal{F}$ such that $\mu(F) = 0$ is known as a *null set* and the measure space is said to be *complete* if all subsets of null sets belong to $\mathcal{F}$.

Any measure space $(X, \mathcal{F}, \mu)$ can be extended to a complete measure space by adding in all subsets of null sets. In this book, we always implicitly assume this extension to a complete measure space has been made.

Example 1.18 (Lebesgue measure) The usual notion of volume of subsets of $\mathbb{R}^d$ gives rise to Lebesgue measure, which we denote Leb , on the Borel sets $\mathcal{B}(\mathbb{R}^d)$. The measure space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \text{Leb})$ is σ -finite and (once subsets of null sets are included) complete.

Integrals are defined for measurable functions. Let $(X, \mathcal{F}, \mu)$ be a measure space and $(Y, \|\cdot\|_Y)$ be a Banach space.

- Definition 1.19** (measurable) (i) A function $u: X \rightarrow \mathbb{R}$ is $\mathcal{F}$ -measurable if $\{x \in X: u(x) \leq a\} \in \mathcal{F}$ for every $a \in \mathbb{R}$. A function $u: X \rightarrow Y$ is $\mathcal{F}$ -measurable if the pullback set $u^{-1}(G) \in \mathcal{F}$ for any $G \in \mathcal{B}(Y)$. We simply write u is measurable if the underlying σ -algebra is clear and write u is Borel measurable if $\mathcal{F}$ is a Borel σ -algebra.
- (ii) A measurable function $u = 0$ almost surely (a.s.) or almost everywhere (a.e.) if $\mu(\{x \in X: u(x) \neq 0\}) = 0$. We sometimes write μ -a.s. to stress the measure.

The following lemma requires the underlying measure space $(X, \mathcal{F}, \mu)$ to be complete (as we assume throughout the book). See [Exercise 1.2](#).

Lemma 1.20 If $u_n: X \rightarrow Y$ is measurable and $u_n(x) \rightarrow u(x)$ for almost all $x \in X$, then $u: X \rightarrow Y$ is measurable.

Finally, we define the integral of a measurable function $u: X \rightarrow Y$ with respect to $(X, \mathcal{F}, \mu)$.

Definition 1.21 (integral)

- (i) A function $s: X \rightarrow Y$ is simple if there exist $s_j \in Y$ and $F_j \in \mathcal{F}$ for $j = 1, \dots, N$ with $\mu(F_j) < \infty$ such that

$$s(x) = \sum_{j=1}^N s_j 1_{F_j}(x), \quad x \in X.$$

The integral of a simple function s with respect to a measure space $(X, \mathcal{F}, \mu)$ is

$$\int_X s(x) d\mu(x) := \sum_{j=1}^N s_j \mu(F_j). \quad (1.3)$$

- (ii) We say a measurable function u is integrable with respect to μ if there exist simple functions u_n such that $u_n(x) \rightarrow u(x)$ as $n \rightarrow \infty$ for almost all $x \in X$ and u_n is a Cauchy sequence in the sense that, for all $\epsilon > 0$,

$$\int_X \|u_n(x) - u_m(x)\|_Y d\mu(x) < \epsilon,$$

for any n, m sufficiently large. Notice that $s(x) = \|u_n(x) - u_m(x)\|_Y$ is a simple function from X to $\mathbb{R}$ and the integral is defined by (1.3).

- (iii) If u is integrable, define

$$\int_X u(x) d\mu(x) := \lim_{n \rightarrow \infty} \int_X u_n(x) d\mu(x).$$

Note that $\int_X \|u(x)\|_Y d\mu(x) < \infty$ (see [Exercise 1.3](#)). If $F \in \mathcal{F}$, define

$$\int_F u(x) d\mu(x) := \int_X u(x) 1_F(x) d\mu(x).$$

When $Y = \mathbb{R}^d$, the above definition gives the Lebesgue integral and, for a Borel set $D \subset \mathbb{R}^d$, the integral with respect to $(D, \mathcal{B}(D), \text{Leb})$ is denoted

$$\int_D u(x) d\text{Leb}(x) = \int_D u(x) dx.$$

This corresponds to the usual notion of volume beneath a surface. However, the Lebesgue integral is much more general than the Riemann integral, as the partitions essential for the definition of the Riemann integral do not make sense for the probability spaces of Chapter 4. [Definition 1.21](#) includes the case where Y is a Banach space, the so-called *Bochner integral*.

The following elementary properties hold in general for integrable functions.

- (i) For integrable functions $u, v: X \rightarrow Y$ and $a, b \in \mathbb{R}$,

$$\int_X (au(x) + bv(x)) d\mu(x) = a \int_X u(x) d\mu(x) + b \int_X v(x) d\mu(x).$$

- (ii) For $u: X \rightarrow Y$,

$$\left\| \int_X u(x) d\mu(x) \right\|_Y \leq \int_X \|u(x)\|_Y d\mu(x). \quad (1.4)$$

We state without proof the dominated convergence theorem on limits of integrals and Fubini's theorem for integration on product spaces.

Theorem 1.22 (dominated convergence) *Consider a sequence of measurable functions $u_n: X \rightarrow Y$ such that $u_n(x) \rightarrow u(x)$ in Y as $n \rightarrow \infty$ for almost all $x \in X$. If there is a real-valued integrable function $\bar{U}$ such that $\|u_n(x)\|_Y \leq |\bar{U}(x)|$ for $n \in \mathbb{N}$ and almost all $x \in X$, then*

$$\lim_{n \rightarrow \infty} \int_X u_n(x) d\mu(x) = \int_X \lim_{n \rightarrow \infty} u_n(x) d\mu(x) = \int_X u(x) d\mu(x).$$

Now suppose that $(X_k, \mathcal{F}_k, \mu_k)$ for $k = 1, 2$ are a pair of σ -finite measure spaces.

Definition 1.23 (product measure) Denote by $\mathcal{F}_1 \times \mathcal{F}_2$ the smallest σ -algebra containing all sets $F_1 \times F_2$ for $F_k \in \mathcal{F}_k$. The *product measure* $\mu_1 \times \mu_2$ on $\mathcal{F}_1 \times \mathcal{F}_2$ is defined by $(\mu_1 \times \mu_2)(F_1 \times F_2) := \mu_1(F_1) \mu_2(F_2)$.

Theorem 1.24 (Fubini) *Suppose that $(X_k, \mathcal{F}_k, \mu_k)$ for $k = 1, 2$ are σ -finite measure spaces and consider a measurable function $u: X_1 \times X_2 \rightarrow Y$. If*

$$\int_{X_2} \left(\int_{X_1} \|u(x_1, x_2)\|_Y d\mu_1(x_1) \right) d\mu_2(x_2) < \infty, \quad (1.5)$$

then u is integrable with respect to the product measure $\mu_1 \times \mu_2$ and

$$\begin{aligned} \int_{X_1 \times X_2} u(x_1, x_2) d(\mu_1 \times \mu_2)(x_1, x_2) &= \int_{X_2} \left(\int_{X_1} u(x_1, x_2) d\mu_1(x_1) \right) d\mu_2(x_2) \\ &= \int_{X_1} \left(\int_{X_2} u(x_1, x_2) d\mu_2(x_2) \right) d\mu_1(x_1). \end{aligned}$$

To understand the necessity of (1.5), see [Exercise 1.7](#).

Example 1.25 For Borel sets $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$, both measure spaces $(X, \mathcal{B}(X), \text{Leb})$ and $(Y, \mathcal{B}(Y), \text{Leb})$ are σ -finite. Consider $u: X \times Y \rightarrow \mathbb{R}$ such that $\int_X \int_Y |u(x, y)| dy dx < \infty$. Then Fubini's theorem applies and

$$\int_{X \times Y} u(x, y) d(\text{Leb} \times \text{Leb})(x, y) = \int_X \int_Y u(x, y) dy dx = \int_Y \int_X u(x, y) dx dy.$$

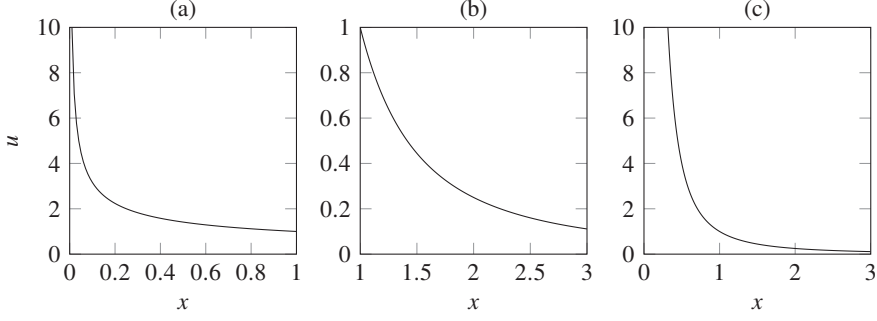


Figure 1.2 (a) If $u(x) = x^{-1/2}$ then $u \in L^p(0, 1)$ for $1 \leq p < 2$; (b) if $u(x) = x^{-2}$ then $u \in L^p(1, \infty)$ for $p > 2$; (c) if $u(x) = x^{-2}$ then $u \notin L^p(0, \infty)$ for any $p \geq 1$.

L^p spaces

With Lebesgue integrals now defined, we introduce spaces of functions that have finite integrals.

Definition 1.26 (L^p spaces) Let $(Y, \|\cdot\|_Y)$ be a Banach space and $1 \leq p < \infty$.

- (i) For a domain D , $L^p(D)$ is the set of Borel measurable functions $u: D \rightarrow \mathbb{R}$ with $\|u\|_{L^p(D)} < \infty$, where

$$\|u\|_{L^p(D)} := \left(\int_D |u(x)|^p dx \right)^{1/p}. \quad (1.6)$$

$L^1(D)$ comprises the real-valued functions on D that are integrable with respect to Lebesgue measure.

- (ii) Let $(X, \mathcal{F}, \mu)$ be a measure space. The space $L^p(X, Y)$ is the set of $\mathcal{F}$ -measurable functions $u: X \rightarrow Y$ such that $\|u\|_{L^p(X, Y)} < \infty$, where

$$\|u\|_{L^p(X, Y)} := \left(\int_X \|u(x)\|_Y^p d\mu(x) \right)^{1/p}.$$

$L^1(X, Y)$ comprises the integrable (with respect to μ) functions $X \rightarrow Y$. We write $L^p(X)$ for $L^p(X, \mathbb{R})$.

- (iii) $L^\infty(X, Y)$ is the set of $\mathcal{F}$ -measurable functions $u: X \rightarrow Y$ such that

$$\|u\|_{L^\infty(X, Y)} := \operatorname{ess\,sup}_{x \in X} \|u(x)\|_Y < \infty$$

and $\operatorname{ess\,sup}_{x \in X} \|u(x)\|_Y$ denotes the essential supremum or the smallest number that bounds $\|u\|_Y$ almost everywhere. We write $L^\infty(X)$ for $L^\infty(X, \mathbb{R})$.

We examine further the case $p = 2$ in §1.2. The following example illustrates $L^p(D)$ for different choices of p and D .

Example 1.27 Let $D = (0, 1)$ and $u(x) = x^\alpha$ with $\alpha < 0$. Then $u \in L^p(D)$ if and only if

$$\int_0^1 x^{\alpha p} dx = \left[\frac{x^{1+\alpha p}}{1+\alpha p} \right]_0^1 < \infty \iff 1 + \alpha p > 0.$$

Hence, $u \in L^p(D)$ for $1 \leq p < -1/\alpha$. See Figure 1.2(a) for the case $\alpha = -1/2$.

If $D = (1, \infty)$, then $u \in L^p(D)$ if and only if $1 + \alpha p < 0$ or $p > -1/\alpha$. See Figure 1.2(b) for the case $\alpha = -2$. If $D = (0, \infty)$ then $u \notin L^p(D)$ for any $p \geq 1$, as u grows too quickly near the origin or decays too slowly at infinity; see Figure 1.2(c).

More generally, it can be shown $L^q(D) \subset L^p(D)$ for $1 \leq p \leq q$ for any bounded domain D . See Exercise 1.9.

The basic inequalities for working on Lebesgue spaces are the following:

Hölder's inequality: Suppose that $1 \leq p, q \leq \infty$ with $1/p + 1/q = 1$ (p, q are said to be *conjugate exponents* and this includes $\{p, q\} = \{1, \infty\}$). For $u \in L^p(X)$ and $v \in L^q(X)$,

$$\|uv\|_{L^1(X)} \leq \|u\|_{L^p(X)} \|v\|_{L^q(X)}. \quad (1.7)$$

Minkowski's inequality: For $u, v \in L^p(X, Y)$,

$$\|u + v\|_{L^p(X, Y)} \leq \|u\|_{L^p(X, Y)} + \|v\|_{L^p(X, Y)}.$$

Minkowski's inequality provides the triangle inequality needed to establish that $\|\cdot\|_{L^p(X, Y)}$ is a norm. Note however that, as functions in $L^p(X, Y)$ in Definition 1.26 are unique only up to sets of measure zero, $\|\cdot\|_{L^p(X, Y)}$ fails the first axiom in Definition 1.1. The trick is to work on the set of equivalence classes of functions that are equal almost everywhere with respect to the given measure on X . Then, when we refer to functions in the $L^p(X, Y)$ spaces, we mean functions that are representative of those equivalence classes. Using a rigorous definition, it can be shown that $L^p(X, Y)$ is a complete normed vector space.

Lemma 1.28 $L^p(X, Y)$ is a Banach space for $1 \leq p \leq \infty$.

1.2 Hilbert spaces L^2 and H^r

Hilbert spaces are Banach spaces with the additional structure of an inner product.

Definition 1.29 (inner product) An *inner product* on a real (resp., complex) vector space X is a function $\langle \cdot, \cdot \rangle: X \times X \rightarrow \mathbb{R}$ (resp., $\mathbb{C}$) that is

- (i) positive definite: $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ if and only if $u = 0$,
- (ii) conjugate symmetric: $\langle u, v \rangle = \overline{\langle v, u \rangle}$, where $\overline{}$ denotes the complex conjugate of u , and
- (iii) linear in the first argument: $\langle \lambda u + \mu v, w \rangle = \lambda \langle u, w \rangle + \mu \langle v, w \rangle$ for real (resp., complex) λ, μ and $u, v, w \in X$.

Definition 1.30 (Hilbert space) Let H be a real (resp., complex) vector space with inner product $\langle \cdot, \cdot \rangle$. Then, H is a real (resp., complex) *Hilbert space* if it is complete with respect to the induced norm $\|u\| := \langle u, u \rangle^{1/2}$. In particular, any Hilbert space is a Banach space.

Example 1.31 $\mathbb{R}^d$ and $\mathbb{C}^d$ are Hilbert spaces with the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^T \bar{\mathbf{y}} = x_1 \bar{y}_1 + x_2 \bar{y}_2 + \cdots + x_d \bar{y}_d.$$

In this book, the norm $\|\cdot\|_2$ on $\mathbb{R}^d$ denotes $\|\mathbf{x}\|_2 := \langle \mathbf{x}, \mathbf{x} \rangle^{1/2} = (x_1^2 + \cdots + x_d^2)^{1/2}$.

The following inequality is frequently used when working with Hilbert spaces.

Lemma 1.32 (Cauchy–Schwarz inequality) *Let H be a Hilbert space. Then*

$$|\langle u, v \rangle| \leq \|u\| \|v\|, \quad \forall u, v \in H. \quad (1.8)$$

Proof See [Exercise 1.10](#). □

Hilbert space $L^2(D)$

Recall the space $L^p(D)$ introduced in Definition 1.26. When $p = 2$, there is an inner product and $L^2(D)$ is a Hilbert space. Functions in this space are called *square integrable*.

Proposition 1.33 *For any domain D , $L^2(D)$ is a real Hilbert space with inner product*

$$\langle u, v \rangle_{L^2(D)} := \int_D u(\mathbf{x})v(\mathbf{x}) d\mathbf{x}.$$

Proof $L^2(D)$ is a Banach space by [Lemma 1.28](#). It is elementary to verify that $\langle \cdot, \cdot \rangle_{L^2(D)}$ is an inner product and that $\langle u, u \rangle_{L^2(D)}^{1/2}$ agrees with (1.6) with $p = 2$. □

More generally, we may work with the L^2 space of functions taking values in any Hilbert space H .

Proposition 1.34 *Let $(X, \mathcal{F}, \mu)$ be a measure space and H be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$. Then, $L^2(X, H)$ is a Hilbert space with inner product*

$$\langle u, v \rangle_{L^2(X, H)} := \int_X \langle u(x), v(x) \rangle d\mu(x).$$

Example 1.35 Later on, we meet the complex Hilbert space $L^2(\mathbb{R}^d, \mathbb{C})$ with inner product $\langle u, v \rangle_{L^2(\mathbb{R}^d, \mathbb{C})} := \int_{\mathbb{R}^d} u(\mathbf{x})\overline{v(\mathbf{x})} d\mathbf{x}$. For stochastic PDEs and a second measurable space $(\Omega, \mathcal{F}, \mathbb{P})$, we use $L^2(\Omega, L^2(D))$, the square integrable functions $u(\mathbf{x}, \omega)$ on $D \times \Omega$ such that $u(\cdot, \omega) \in L^2(D)$ for almost every $\omega \in D$, with inner product

$$\langle u, v \rangle_{L^2(\Omega, L^2(D))} := \int_{\Omega} \int_D u(\mathbf{x}, \omega)\overline{v(\mathbf{x}, \omega)} d\mathbf{x} d\mathbb{P}(\omega).$$

Orthogonal projections and orthonormal bases

Consider a Hilbert space H with inner product $\langle \cdot, \cdot \rangle$ and let G be a subspace of H (i.e., $G \subset H$ and is a vector space).

Definition 1.36 (orthogonal complement) *The orthogonal complement of a closed subspace G of H is*

$$G^\perp := \{u \in H : \langle u, v \rangle = 0 \text{ for all } v \in G\}.$$

Note that $G \cap G^\perp = \{0\}$. For a non-zero vector $\mathbf{u} \in \mathbb{R}^d$, $G = \text{span}\{\mathbf{u}\}$ is a subspace of $\mathbb{R}^d$ and $G^\perp$ is the hyperplane through the origin orthogonal to $\mathbf{u}$.

Theorem 1.37 *If G is a closed subspace of a Hilbert space H , every $u \in H$ may be uniquely written $u = p^* + q$ for $p^* \in G$ and $q \in G^\perp$. Further, $\|u - p^*\| = \inf_{p \in G} \|u - p\|$.*

Proof We assume that H is a real Hilbert space. Consider $p_n \in G$ such that $\|u - p_n\| \rightarrow \inf_{p \in G} \|u - p\|$ as $n \rightarrow \infty$. As G is closed, it may be shown that p_n is a Cauchy sequence that has a limit point p^* in G (see Exercise 1.16). Clearly $\|u - p^*\| = \inf_{p \in G} \|u - p\|$. For $v \in G$, let $\phi(\epsilon) := \|u - p^* + \epsilon v\|^2$. Then $\phi(\epsilon)$ has a minimum at $\epsilon = 0$ and $\phi'(0) = 2\langle u - p^*, v \rangle = 0$ for all $v \in G$, which means that $u - p^* \in G^\perp$. To show uniqueness, suppose $u = p_i + q_i$ for $i = 1, 2$ with $p_i \in G$, $q_i \in G^\perp$. Because $G \cap G^\perp = \{0\}$, we see that $p_1 - p_2 = q_1 - q_2 = 0$ and the decomposition is unique. $\square$

Definition 1.38 (orthogonal projection) The orthogonal projection $P: H \rightarrow G$ is defined by $Pu = p^*$, where p^* is given in Theorem 1.37. Note that $P^2 = P$.

We can represent the orthogonal projection using an orthonormal basis of H .

Definition 1.39 (orthonormal basis) Let H be a Hilbert space. A set $\{\phi_j: j \in \mathbb{N}\} \subset H$ is *orthonormal* if

$$\langle \phi_j, \phi_k \rangle = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

This is also written as $\langle \phi_j, \phi_k \rangle = \delta_{jk}$, where δ_{jk} is known as the *Kronecker delta* function. If the span of $\{\phi_j\}$ is additionally dense in H then we have a *complete* orthonormal set and $\{\phi_j\}$ is said to be an *orthonormal basis* for H . In that case, $u = \sum_{j=1}^{\infty} \langle u, \phi_j \rangle \phi_j$ for any $u \in H$ and

$$\|u\|^2 = \sum_{j=1}^{\infty} |\langle u, \phi_j \rangle|^2 = \sum_{j=1}^{\infty} |\langle \phi_j, u \rangle|^2. \quad (1.9)$$

A Hilbert space is *separable* if it contains a countable dense subset and every separable Hilbert space has an orthonormal basis (as can be proved by applying the Gram–Schmidt orthogonalisation procedure to the countable dense subset).

Example 1.40 For any domain D , $L^2(D)$ is separable and has an orthonormal basis. For example,

$$\begin{aligned} \left\{ \sqrt{2} \sin(j\pi x) : j \in \mathbb{N} \right\} & \text{ is an orthonormal basis for } L^2(0, 1), \\ \left\{ \frac{1}{\sqrt{2\pi}} e^{ikx} : k \in \mathbb{Z} \right\} & \text{ is an orthonormal basis for } L^2((-\pi, \pi), \mathbb{C}). \end{aligned}$$

We frequently work with the orthogonal projection from H to the span of the first N basis functions.

Lemma 1.41 Consider a separable Hilbert space H with an orthonormal basis $\{\phi_j: j \in \mathbb{N}\}$. Let $G_N := \text{span}\{\phi_1, \dots, \phi_N\}$ and denote the orthogonal projection from H to G_N by P_N . Then,

$$P_N u = \sum_{j=1}^N \langle u, \phi_j \rangle \phi_j. \quad (1.10)$$

In particular, $\|P_N u\| \leq \|u\|$ and $\|u - P_N u\| \rightarrow 0$ as $N \rightarrow \infty$.

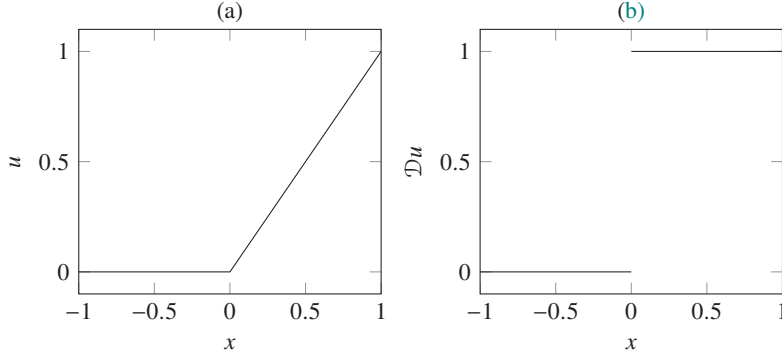


Figure 1.3 (a) Plot of $u(x)$ defined by (1.11) and (b) the derivative of $u(x)$ defined for $x \neq 0$.

Proof Let $u \in H$. For any $p \in G_N$,

$$u - p = \sum_{j=1}^N \langle u - p, \phi_j \rangle \phi_j + \sum_{j=N+1}^{\infty} \langle u, \phi_j \rangle \phi_j$$

and $\|u - p\|^2 = \sum_{j=1}^N |\langle u - p, \phi_j \rangle|^2 + \sum_{j=N+1}^{\infty} |\langle u, \phi_j \rangle|^2$. Hence, $\|u - p\|$ is minimized by choosing $p = \sum_{j=1}^N \langle u, \phi_j \rangle \phi_j$ and the orthogonal projection P_N is given by (1.10). We have $\|P_N u\| \leq \|u\|$ because

$$\|u\|^2 = \sum_{j=1}^{\infty} |\langle \phi_j, u \rangle|^2 \quad \text{and} \quad \|P_N u\|^2 = \sum_{j=1}^N |\langle \phi_j, u \rangle|^2.$$

Finally, $\|u - P_N u\|^2 = \sum_{j=N+1}^{\infty} |\langle \phi_j, u \rangle|^2 \rightarrow 0$ as $N \rightarrow \infty$. $\square$

Sobolev spaces

Just as $C^r(D, Y)$ describes the regularity of continuous functions, Sobolev spaces describe the regularity of Lebesgue-integrable functions. We assign a meaning to the derivative $\mathcal{D}u$, even when u may not have a derivative in the traditional sense. For example,

$$u(x) = \begin{cases} 0, & x < 0, \\ x, & x \geq 0 \end{cases} \quad (1.11)$$

is piecewise differentiable but, at $x = 0$, the derivative is not defined. See Figure 1.3. We define the derivative

$$\mathcal{D}u(x) = \begin{cases} 0, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0, \end{cases} \quad (1.12)$$

where we have made a choice for $\mathcal{D}u(0)$. To make a formal definition of the derivative and show that this choice for $\mathcal{D}u$ makes sense, we integrate $\mathcal{D}u$ by parts against a test function in $C_c^\infty(D)$ (see Definition 1.13). The space $C_c^\infty(D)$ is used because the functions

are differentiable and the space is dense in $L^2(D)$ (any function in $L^2(D)$ is approximated arbitrarily well by one in $C_c^\infty(D)$). Let $D = (a, b)$ and $\phi \in C_c^\infty(D)$. Integration by parts gives

$$\int_a^b \mathcal{D}u(x)\phi(x) dx = \left[u(x)\phi(x) \right]_a^b - \int_a^b u(x)\mathcal{D}\phi(x) dx.$$

Because ϕ has support in D , ϕ is identically zero outside D and ϕ and all its derivatives are zero on the boundary of D . In particular, $\phi(a) = \phi(b) = 0$ and

$$\int_a^b \mathcal{D}u(x)\phi(x) dx = - \int_a^b u(x)\mathcal{D}\phi(x) dx, \quad \forall \phi \in C_c^\infty(\mathbb{R}). \quad (1.13)$$

It is easy to show that (1.12) satisfies this condition and, because the integral is independent of the choice for $\mathcal{D}u(0)$, it does not matter what value is chosen for $\mathcal{D}u(0)$ in (1.12) (see Exercise 1.11). Repeatedly integrating by parts, for $\alpha \in \mathbb{N}$,

$$\int_a^b \mathcal{D}^\alpha u(x)\phi(x) dx = (-1)^\alpha \int_a^b u(x)\mathcal{D}^\alpha \phi(x) dx.$$

This calculation for functions u of $x \in \mathbb{R}$ also extends to functions of $\mathbf{x} \in \mathbb{R}^d$ by using the multi-index notation.

Definition 1.42 (weak derivative) Let Y be a Banach space. We say a measurable function $\mathcal{D}^\alpha u: D \rightarrow Y$ is the α th weak derivative of a measurable function $u: D \rightarrow Y$ if

$$\int_D \mathcal{D}^\alpha u(\mathbf{x})\phi(\mathbf{x}) d\mathbf{x} = (-1)^{|\alpha|} \int_D u(\mathbf{x})\mathcal{D}^\alpha \phi(\mathbf{x}) d\mathbf{x}, \quad \forall \phi \in C_c^\infty(D).$$

Example 1.43 (Dirac delta function) $\mathcal{D}u$ defined by (1.12) is constant for $x \neq 0$ and hence $\mathcal{D}^2 u(x)$, the derivative of $\mathcal{D}u$, is zero for all $x \neq 0$. However, the zero function in $L^2(\mathbb{R})$ is not the derivative of $\mathcal{D}u$. In fact, $\mathcal{D}^2 u(x) = \delta(x)$, the Dirac delta function, defined by

$$\delta(x) = 0 \quad \text{for } x \neq 0, \quad \int_{\mathbb{R}} \delta(x)\phi(x) dx = \phi(0),$$

for any continuous function $\phi(x)$. In particular, $\int_{-\epsilon}^{\epsilon} \delta(x) dx = 1$ for any $\epsilon > 0$ and the delta function has unit mass at $x = 0$, representing the derivative of $\mathcal{D}u$ at $x = 0$. The delta function is frequently used, even though it is not a function in the classical sense and, strictly speaking, requires the theory of distributions.

Definition 1.44 (Sobolev spaces) Let D be a domain and Y be a Banach space. For $p \geq 1$, the Sobolev space $W^{r,p}(D, Y)$ is the set of functions whose weak derivatives up to order $r \in \mathbb{N}$ are in $L^p(D, Y)$. That is,

$$W^{r,p}(D, Y) := \{u: \mathcal{D}^\alpha u \in L^p(D, Y) \text{ if } |\alpha| \leq r\}.$$

If $p = 2$ and H is a Hilbert space, $H^r(D, H)$ is used to denote $W^{r,2}(D, H)$.

Proposition 1.45 $W^{r,p}(D, Y)$ is a Banach space with norm

$$\|u\|_{W^{r,p}(D, Y)} := \left(\sum_{0 \leq |\alpha| \leq r} \|\mathcal{D}^\alpha u\|_{L^p(D, Y)}^p \right)^{1/p}$$

and $H^r(D, H)$ is a Hilbert space with inner product

$$\langle u, v \rangle_{H^r(D, H)} := \sum_{0 \leq |\alpha| \leq r} \langle \mathcal{D}^\alpha u, \mathcal{D}^\alpha v \rangle_{L^2(D, H)}.$$

We abbreviate $H^r(D, \mathbb{R})$ to $H^r(D)$ and the norm on $H^r(D)$ is

$$\|u\|_{H^r(D)} := \left(\sum_{0 \leq |\alpha| \leq r} \|\mathcal{D}^\alpha u\|_{L^2(D)}^2 \right)^{1/2}. \quad (1.14)$$

It is also convenient to define the Sobolev semi-norm (recall Definition 1.1):

$$|u|_{H^r(D)} := \left(\sum_{|\alpha|=r} \|\mathcal{D}^\alpha u\|_{L^2(D)}^2 \right)^{1/2}. \quad (1.15)$$

By using fractional powers (see Definition 1.87), it is possible to define fractional derivatives and define Sobolev spaces for any $r \in \mathbb{R}$.

Sobolev spaces that incorporate boundary conditions are needed to analyse solutions to boundary-value problems. For one-dimensional domains $D = (a, b)$, the most frequently used in our studies are, for homogeneous Dirichlet conditions,

$$H_0^1(a, b) := \{u \in H^1(a, b) : u(a) = u(b) = 0\},$$

and, for periodic conditions,

$$H_{\text{per}}^2(a, b) := \{u \in H^2(a, b) : u(a) = u(b) \text{ and } u_x(a) = u_x(b)\}.$$

Precise definitions are given below.

Definition 1.46 (completion) X is the completion of Y with respect to $\|\cdot\|$ if X equals the union of the space Y and the limit points of sequences in Y with respect to $\|\cdot\|$.

Definition 1.47 (Sobolev spaces H_0^1, H_{per}^2)

- (i) For a domain D , $H_0^1(D)$ is the completion of $C_c^\infty(D)$ with respect to the $H^1(D)$ norm and is a Hilbert space with the $H^1(D)$ inner product.
- (ii) $H_{\text{per}}^2(a, b)$ is the completion with respect to the $H^2(a, b)$ norm of the set of $u \in C^\infty([a, b])$ such that the p th derivative $u^{(p)}(a) = u^{(p)}(b)$ for $p = 0, 1, \dots$. It is a Hilbert space with the $H^2(D)$ inner product.
- (iii) Let $D := (a_1, b_1) \times (a_2, b_2) \subset \mathbb{R}^2$ be a domain. $H_{\text{per}}^2(D)$ is the completion with respect to the $H^2(D)$ norm of the set of $u \in C^\infty(\bar{D})$ such that

$$\frac{\partial^p}{\partial x_1^p} u(a_1, y) = \frac{\partial^p}{\partial x_1^p} u(b_1, y), \quad \frac{\partial^p}{\partial x_2^p} u(x, a_2) = \frac{\partial^p}{\partial x_2^p} u(x, b_2)$$

for $x \in (a_1, b_1)$, $y \in (a_2, b_2)$, and $p = 0, 1, \dots$. It is a Hilbert space with the $H^2(D)$ inner product.

Below, we introduce a norm on $H_0^1(D)$ that is equivalent to $\|\cdot\|_{H^1(D)}$. First, we state Poincaré's inequality. Recall the definition of $|u|_{H^1(D)}$ from (1.15).

Theorem 1.48 (Poincaré's inequality) *For a bounded domain D , there exists a constant $K_p > 0$ such that $\|u\|_{L^2(D)} \leq K_p |u|_{H^1(D)}$ for any $u \in H_0^1(D)$.*

Proof See [Exercise 1.14](#) for the case $D \subset \mathbb{R}$. $\square$

Definition 1.49 (equivalence of norms) Consider two norms $\|\cdot\|_A$ and $\|\cdot\|_B$ on a Banach space X . We say the norms are *equivalent* if there exist constants $K_1, K_2 > 0$ such that

$$K_1 \|u\|_A \leq \|u\|_B \leq K_2 \|u\|_A, \quad \forall u \in X.$$

For equivalent norms, a sequence u_n converges to u with respect to $\|\cdot\|_A$ or equivalently $\|u_n - u\|_A \rightarrow 0$ as $n \rightarrow \infty$ if and only if $\|u_n - u\|_B \rightarrow 0$ as $n \rightarrow \infty$.

Lemma 1.50 Suppose that D is a bounded domain. Then $\|u\|_{H_0^1(D)} := |u|_{H^1(D)}$ is a norm. Further, $\|u\|_{H^1(D)}$ and $\|u\|_{H_0^1(D)}$ are equivalent norms on $H_0^1(D)$. In particular,

$$\|u\|_{H_0^1(D)} \leq \|u\|_{H^1(D)} \leq (1 + K_p^2)^{1/2} \|u\|_{H_0^1(D)}, \quad \forall u \in H_0^1(D).$$

Proof This is a simple consequence of the definitions of $\|\cdot\|_{H^1(D)}$ and $\|\cdot\|_{H_0^1(D)}$ and the Poincaré inequality ([Theorem 1.48](#)). $\square$

The relationship between the space $H^r(D)$ of functions with Lebesgue-integrable weak derivatives and the space $C^r(\bar{D})$ of functions with classical derivatives is described by the Sobolev embedding theorems.

Theorem 1.51 (Sobolev embedding) Let D be a bounded domain in $\mathbb{R}^d$ and $u \in H^r(D)$ for $r \in \mathbb{N}$. If $r > d/2$, then $u: D \rightarrow \mathbb{R}$ can be extended to a continuous function $u: \bar{D} \rightarrow \mathbb{R}$. Further, $u \in C(\bar{D})$ and there exists $K > 0$ such that $\|u\|_{C(\bar{D})} \leq K \|u\|_{H^r(D)}$.

If $r > s + d/2$ for $s \in \mathbb{N}$, then $u \in C^s(\bar{D})$ and there exists $K > 0$ such that $\|u\|_{C^s(\bar{D})} \leq K \|u\|_{H^r(D)}$.

The theorem is dimension dependent. Any function in $H^1(D)$ for $D = (a, b)$ is continuous and bounded, but in higher dimensions this is not true. For example in dimension $d = 2$, the function $u(\mathbf{x}) = \log|\log r|$, for $r = \|\mathbf{x}\|_2$, belongs to $H^1(D)$ for $D = \{\mathbf{x} \in \mathbb{R}^2: \|\mathbf{x}\|_2 < 1/2\}$. However, $u(\mathbf{x})$ is unbounded at $\mathbf{x} = \mathbf{0}$ and does not belong to $C(\bar{D})$. See [Exercise 1.12](#).

1.3 Linear operators and spectral theory

Definition 1.52 (linear operator) Let X and Y be two real (resp., complex) vector spaces. A function $L: X \rightarrow Y$ is said to be a *linear operator* if, for any $u, v \in X$ and any scalar $a \in \mathbb{R}$ (resp., $\mathbb{C}$),

$$L(u + v) = L(u) + L(v), \quad L(av) = aL(v).$$

We usually drop the brackets and write Lu for $L(u)$.

Examples include matrix multiplication of vectors in $\mathbb{R}^d$, the projection operator P_N of [Lemma 1.41](#), and differentiation of functions in $C^r(D)$ or $H^r(D)$. We first discuss operators L that are bounded and compact, before looking at the Hilbert–Schmidt spectral theory.

Definition 1.53 (bounded linear operator) Let $(X, \|\cdot\|_X), (Y, \|\cdot\|_Y)$ be Banach spaces. A linear operator $L: X \rightarrow Y$ is *bounded* if for some $K > 0$

$$\|Lu\|_Y \leq K \|u\|_X, \quad \forall u \in X.$$

We write $\mathcal{L}(X, Y)$ for the set of bounded linear operators $L: X \rightarrow Y$. In the case $X = Y$, we write $\mathcal{L}(X) = \mathcal{L}(X, X)$.

Bounded linear operators are important because they are exactly the continuous linear operators. (see [Exercise 1.15](#)).

Lemma 1.54 *The set $\mathcal{L}(X, Y)$ is a Banach space with the operator norm*

$$\|L\|_{\mathcal{L}(X, Y)} := \sup_{u \neq 0} \frac{\|Lu\|_Y}{\|u\|_X}.$$

If $L \in \mathcal{L}(X, Y)$, then $\|Lu\|_Y \leq \|L\|_{\mathcal{L}(X, Y)} \|u\|_X$ for any $u \in X$.

Proof See [Exercise 1.15](#). □

Example 1.55 (matrices) With $X = \mathbb{R}^d = Y$, multiplication by a $d \times d$ matrix is a bounded linear operation. We characterise the $\mathcal{L}(\mathbb{R}^d)$ norm of symmetric matrices in [Example 1.71](#).

Again, the case $Y = \mathbb{R}$ deserves special attention.

Definition 1.56 (bounded linear functional) A *bounded linear functional* on X is a member of $\mathcal{L}(X, \mathbb{R})$. The set of bounded linear functionals on a Banach space is known as the *dual space* and is itself a Banach space with the norm $\|\cdot\|_{\mathcal{L}(X, \mathbb{R})}$.

Bounded linear functionals are characterised by the Riesz representation theorem and the Lax–Milgram lemma.

Theorem 1.57 (Riesz representation) *Let H be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and let ℓ be a bounded linear functional on H . There exists a unique $u_\ell \in H$ such that*

$$\langle u_\ell, x \rangle = \ell(x), \quad \forall x \in H.$$

Definition 1.58 (bilinear form) Let X, Y and Z be three real vector spaces. A function $a: X \times Y \rightarrow Z$ is a *bilinear form* if, for any $y \in Y$, $x \mapsto a(x, y)$ is a linear operator from X to Z and, for any $x \in X$, $y \mapsto a(x, y)$ is a linear operator from Y to Z .

If X, Y, Z have norms $\|\cdot\|_X$, $\|\cdot\|_Y$, and $\|\cdot\|_Z$, we say a is *bounded* if, for some $\alpha > 0$,

$$\|a(x, y)\|_Z \leq \alpha \|x\|_X \|y\|_Y, \quad \forall x \in X, y \in Y.$$

If $a: X \times X \rightarrow \mathbb{R}$, we say a is *coercive* if, for some $\beta > 0$,

$$a(x, x) \geq \beta \|x\|_X^2, \quad \forall x \in X.$$

Lemma 1.59 (Lax–Milgram) *Let H be a real Hilbert space with norm $\|\cdot\|$ and let ℓ be a bounded linear functional on H . Let $a: H \times H \rightarrow \mathbb{R}$ be a bilinear form that is bounded and coercive. There exists a unique $u_\ell \in H$ such that $a(u_\ell, x) = \ell(x)$ for all $x \in H$.*

The Lax–Milgram lemma implies the Riesz representation theorem, by using the inner product $\langle \cdot, \cdot \rangle$ on H for a . Lax–Milgram, however, is more general because a need not be symmetric or an inner product. Both theorems provide existence and uniqueness of a u_ℓ that represents the bounded linear functional ℓ and are frequently used for the proof of existence and uniqueness of solutions to PDEs (see [Chapter 2](#)).

Hilbert–Schmidt operators

An important class of bounded linear operators are the Hilbert–Schmidt operators.

Definition 1.60 Let H, U be separable Hilbert spaces with norms $\|\cdot\|, \|\cdot\|_U$ respectively. For an orthonormal basis $\{\phi_j : j \in \mathbb{N}\}$ of U , define the Hilbert–Schmidt norm

$$\|L\|_{\text{HS}(U, H)} := \left(\sum_{j=1}^{\infty} \|L\phi_j\|_U^2 \right)^{1/2}.$$

The set $\text{HS}(U, H) := \{L \in \mathcal{L}(U, H) : \|L\|_{\text{HS}(U, H)} < \infty\}$ is a Banach space with the Hilbert–Schmidt norm. An $L \in \text{HS}(U, H)$ is known as a *Hilbert–Schmidt operator*. We write $\|L\|_{\text{HS}} = \|L\|_{\text{HS}(H, H)}$ if $U = H$.

Example 1.61 (matrix) The linear operators from $\mathbb{R}^d$ to $\mathbb{R}^d$ are the $\mathbb{R}^{d \times d}$ matrices, $A = (a_{ij})$. Denote by e_j the standard basis of $\mathbb{R}^d$. The Hilbert–Schmidt norm is $\|A\|_{\text{HS}} = \left(\sum_{i,j=1}^d a_{ij}^2 \right)^{1/2}$ and coincides with the Frobenius norm $\|A\|_F := \left(\sum_{i,j=1}^d a_{ij}^2 \right)^{1/2}$.

Example 1.62 Define $L \in \mathcal{L}(L^2(0, 1))$ by

$$(Lu)(x) = \int_0^x u(y) dy, \quad u \in L^2(0, 1), \quad 0 < x < 1. \quad (1.16)$$

Consider the orthonormal basis $\phi_j(x) = \sqrt{2} \sin(j\pi x)$. We have

$$L\phi_j(x) = \sqrt{2}(1 - \cos(j\pi x)) \frac{1}{j\pi}.$$

Therefore, $\|L\phi_j\|_{L^2(0,1)} \leq 2\sqrt{2}/j\pi$ and L is a Hilbert–Schmidt operator because

$$\|L\|_{\text{HS}}^2 = \sum_{j=1}^{\infty} \|L\phi_j\|_{L^2(0,1)}^2 \leq \sum_{j=1}^{\infty} \frac{8}{j^2\pi^2} < \infty.$$

Lemma 1.63 Let H be separable Hilbert space. If $L \in \text{HS}(H, H)$, then $\|L\|_{\mathcal{L}(H)} \leq \|L\|_{\text{HS}}$. In particular, Hilbert–Schmidt operators are bounded.

Proof Let $\{\phi_j : j \in \mathbb{N}\}$ be an orthonormal basis for H and write $u = \sum_{j=1}^{\infty} u_j \phi_j$ for $u \in H$. Then, $\|u\| = \left(\sum_{j=1}^{\infty} u_j^2 \right)^{1/2}$ and, using the Cauchy–Schwarz inequality,

$$\begin{aligned} \|L\|_{\mathcal{L}(H)} &= \sup_{\|u\|=1} \|Lu\| = \sup_{\|u\|=1} \left\| \sum_{j=1}^{\infty} Lu_j \phi_j \right\| \leq \sup_{\sum_{j=1}^{\infty} |u_j|^2 = 1} \sum_{j=1}^{\infty} |u_j| \|L\phi_j\| \\ &\leq \left(\sum_{j=1}^{\infty} \|L\phi_j\|^2 \right)^{1/2} = \|L\|_{\text{HS}}. \quad \square \end{aligned}$$

We now focus on integral operators and their relationship with the Hilbert–Schmidt operators on $L^2(D)$.

Definition 1.64 (integral operator with kernel G) For a domain D and $G \in L^2(D \times D)$, the integral operator L on $L^2(D)$ with kernel G is defined by

$$(Lu)(\mathbf{x}) := \int_D G(\mathbf{x}, \mathbf{y})u(\mathbf{y}) d\mathbf{y}, \quad \mathbf{x} \in D, \quad u \in L^2(D). \quad (1.17)$$

For example, the operator L in (1.16) is an integral operator on $L^2(0, 1)$ with kernel $G(x, y) = 1_{(0, x)}(y)$.

Theorem 1.65 (integral operators, Hilbert–Schmidt) Any integral operator with kernel $G \in L^2(D \times D)$ is a Hilbert–Schmidt operator on $L^2(D)$.

Conversely, any Hilbert–Schmidt operator L on $L^2(D)$ can be written as (1.17) with $\|L\|_{\text{HS}} = \|G\|_{L^2(D \times D)}$.

Proof First, we show that L defined by (1.17) is Hilbert–Schmidt. $L^2(D)$ is a separable Hilbert space and has an orthonormal basis $\{\phi_j\}$. Further, $\{\phi_j(\mathbf{x})\phi_k(\mathbf{y}) : j, k = 1, \dots, \infty\}$ is an orthonormal basis for $L^2(D \times D)$ and

$$G(\mathbf{x}, \mathbf{y}) = \sum_{j,k=1}^{\infty} g_{jk} \phi_j(\mathbf{x}) \phi_k(\mathbf{y}) \quad (1.18)$$

with coefficients $g_{jk} \in \mathbb{R}$ defined by

$$g_{jk} := \langle G, \phi_j \phi_k \rangle_{L^2(D \times D)} = \int_D \int_D G(\mathbf{x}, \mathbf{y}) \phi_j(\mathbf{x}) \phi_k(\mathbf{y}) d\mathbf{x} d\mathbf{y} = \langle \phi_j, L\phi_k \rangle_{L^2(D)}.$$

Using (1.9), we obtain

$$\|G\|_{L^2(D \times D)} = \left(\sum_{j,k=1}^{\infty} |g_{jk}|^2 \right)^{1/2}, \quad \|L\phi_k\| = \left(\sum_{j=1}^{\infty} |\langle \phi_j, L\phi_k \rangle|^2 \right)^{1/2} \quad (1.19)$$

and so

$$\begin{aligned} \|L\|_{\text{HS}} &= \left(\sum_{k=1}^{\infty} \|L\phi_k\|^2 \right)^{1/2} = \left(\sum_{j,k=1}^{\infty} |\langle \phi_j, L\phi_k \rangle|^2 \right)^{1/2} = \left(\sum_{j,k=1}^{\infty} |g_{jk}|^2 \right)^{1/2} \\ &= \|G\|_{L^2(D \times D)}. \end{aligned} \quad (1.20)$$

Hence, the integral operator L with kernel G is a Hilbert–Schmidt operator with norm $\|L\|_{\text{HS}} = \|G\|_{L^2(D \times D)} < \infty$.

Hilbert–Schmidt operators on $L^2(D)$ can be written as integral operators, as we now show. Write $L\phi_k = \sum_{j=1}^{\infty} g_{jk} \phi_j$ for $g_{jk} = \langle L\phi_k, \phi_j \rangle_{L^2(D)}$. For $u = \sum_{k=1}^{\infty} u_k \phi_k$,

$$Lu = L \sum_{k=1}^{\infty} u_k \phi_k = \sum_{k=1}^{\infty} u_k L\phi_k = \sum_{k=1}^{\infty} u_k \sum_{j=1}^{\infty} \langle L\phi_k, \phi_j \rangle_{L^2(D)} \phi_j = \sum_{j,k=1}^{\infty} g_{jk} \phi_j u_k.$$

Now, substitute $u_k = \langle u, \phi_k \rangle_{L^2(D)}$, to find

$$(Lu)(\mathbf{x}) = \int_D \sum_{j,k=1}^{\infty} g_{jk} \phi_j(\mathbf{x}) \phi_k(\mathbf{y}) u(\mathbf{y}) d\mathbf{y},$$

so that L is an integral operator with kernel G defined by (1.18). As L is a Hilbert–Schmidt operator and (1.20) holds, $\sum_{j,k=1}^{\infty} |g_{jk}|^2 < \infty$ and G is well defined in $L^2(D \times D)$. $\square$

We mention one more property of the integral operators.

Definition 1.66 (compact set) A set B in a Banach space X is *compact* if every sequence u_n in B has a subsequence $\tilde{u}_n$ converging to a limit $u \in B$.

The easiest sets to work with in finite dimensions are the closed and bounded sets, because they are compact (the Heine–Borel theorem). Closed and bounded sets are not compact in infinite dimensions; for example, the sequence defined by an orthonormal basis of a Hilbert space has no convergent subsequences (see [Exercise 1.18](#)). However, compact sets are still easy to find and important, because the integral operators map bounded sets to compact ones.

Definition 1.67 (compact linear operator) A linear operator $L: X \rightarrow Y$, where X and Y are Banach spaces, is *compact* if the image of any bounded set B in X has compact closure in Y . In other words, the closure of $L(B) = \{Lu: u \in B\}$ is compact in Y .

Theorem 1.68 For $G \in L^2(D \times D)$, the integral operator L on $L^2(D)$ defined by (1.17) is *compact*.

Spectral theory of compact symmetric operators

Any real symmetric matrix A can be transformed into a diagonal matrix $\Lambda = Q^T A Q$, by an orthogonal matrix Q of eigenvectors. We now look at the generalisation of this result to infinite-dimensional Hilbert spaces.

Definition 1.69 (symmetric operator) $L \in \mathcal{L}(H)$ is *symmetric* on a Hilbert space H if

$$\langle Lu, v \rangle = \langle u, Lv \rangle \quad \text{for any } u, v \in H.$$

For the Hilbert space $\mathbb{R}^d$ with inner product $\langle u, v \rangle = u^T v$, the symmetric linear operators in $\mathcal{L}(\mathbb{R}^d, \mathbb{R}^d)$ are the $d \times d$ symmetric matrices.

Definition 1.70 (eigenvalue) If $L \in \mathcal{L}(H)$, λ is an *eigenvalue* of L if there exists a non-zero $\phi \in H$ such that $L\phi = \lambda\phi$. Here, ϕ is called an *eigenvector* or *eigenfunction*.

Example 1.71 ($\mathcal{L}(\mathbb{R}^d)$ norm) Let A be a $d \times d$ matrix and recall that

$$\|A\|_{\mathcal{L}(\mathbb{R}^d)} := \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2},$$

where we treat $\mathbb{R}^d$ as the Hilbert space with norm $\|\cdot\|_2$. By differentiating

$$\frac{\|Ax\|_2^2}{\|x\|_2^2} = \frac{x^T A^T A x}{x^T x}$$

with respect to x , we find that the maximum occurs at an eigenvector of $A^T A$. Further, $\|Au\|_2^2 / \|u\|_2^2 = \lambda$, for an eigenvector u of $A^T A$ with eigenvalue λ . Hence $\|A\|_{\mathcal{L}(\mathbb{R}^d)} = (\rho(A^T A))^{1/2}$, where $\rho(\cdot)$ denotes the spectral radius (the magnitude of the largest eigenvalue). When A is symmetric, we obtain $\|A\|_{\mathcal{L}(\mathbb{R}^d)} = \rho(A)$.

Lemma 1.72 For a domain D , suppose that $G \in L^2(D \times D)$ is symmetric (i.e., $G(x, y) = G(y, x)$ for all $x, y \in D$). The integral operator L in (1.17) with kernel G is symmetric.

Proof Let L denote the integral operator defined in (1.17). If $u, v \in L^2(D)$, then $|u|, |v| \in L^2(D)$. Further, $|G| \in L^2(D \times D)$ and hence

$$\int_D \int_D |u(\mathbf{x})G(\mathbf{x}, \mathbf{y})v(\mathbf{y})| d\mathbf{x} d\mathbf{y} < \infty.$$

Therefore, Fubini's theorem applies and

$$\begin{aligned} \langle u, Lv \rangle_{L^2(D)} &= \int_D u(\mathbf{x}) \left(\int_D G(\mathbf{x}, \mathbf{y})v(\mathbf{y}) d\mathbf{y} \right) d\mathbf{x} \\ &= \int_D \left(\int_D G(\mathbf{y}, \mathbf{x})u(\mathbf{x}) d\mathbf{x} \right) v(\mathbf{y}) d\mathbf{y} = \langle Lu, v \rangle_{L^2(D)}. \end{aligned} \quad \square$$

The major theorem concerning eigenvalues of compact symmetric operators, such as integral operators on $H = L^2(D)$, is the Hilbert–Schmidt spectral theorem.

Theorem 1.73 (Hilbert–Schmidt spectral) *Let H be a countably infinite-dimensional Hilbert space and let $L \in \mathcal{L}(H)$ be symmetric and compact. Denote the eigenvalues of L by λ_j , ordered so that $|\lambda_j| \geq |\lambda_{j+1}|$, and denote the corresponding eigenfunctions by $\phi_j \in H$. Then,*

- (i) *all eigenvalues λ_j are real and $\lambda_j \rightarrow 0$ as $j \rightarrow \infty$,*
- (ii) *ϕ_j can be chosen to form an orthonormal basis for the range of L , and*
- (iii) *for any $u \in H$,*

$$Lu = \sum_{j=1}^{\infty} \lambda_j \langle u, \phi_j \rangle \phi_j. \quad (1.21)$$

Example 1.74 (symmetric matrix) If A is a $d \times d$ symmetric matrix, there exists an orthogonal matrix Q of eigenvectors and a diagonal matrix Λ of eigenvalues such that $AQ = Q\Lambda$. Denote the j th entry of Λ by λ_j and the j th column of Q by ϕ_j . We may assume that $|\lambda_{j+1}| \leq |\lambda_j|$. As $\{\phi_j\}$ form an orthonormal basis for $\mathbb{R}^d$, $\mathbf{u} = \sum_{j=1}^d \langle \mathbf{u}, \phi_j \rangle \phi_j$ for any $\mathbf{u} \in \mathbb{R}^d$ and the finite-dimensional analogue of (1.21) is

$$A\mathbf{u} = \sum_{j=1}^d \lambda_j \langle \mathbf{u}, \phi_j \rangle \phi_j.$$

Non-negative definite operators

Definition 1.75 (non-negative definite, positive definite) A function $G: D \times D \rightarrow \mathbb{R}$ is *non-negative definite* if, for any $\mathbf{x}_j \in D$ and $a_j \in \mathbb{R}$ for $j = 1, \dots, N$, we have

$$\sum_{j,k=1}^N a_j a_k G(\mathbf{x}_j, \mathbf{x}_k) \geq 0.$$

A linear operator $L \in \mathcal{L}(H)$ on a Hilbert space H is *non-negative definite* (sometimes called positive semi-definite) if $\langle u, Lu \rangle \geq 0$ for any $u \in H$ and *positive definite* if $\langle u, Lu \rangle > 0$ for any $u \in H \setminus \{0\}$.

Example 1.76 (matrices) Let A be a $d \times d$ real symmetric matrix. If $\mathbf{u}^\top A \mathbf{u} > 0$ (respectively, ≥ 0) for all $\mathbf{u} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$ or, equivalently, if all the eigenvalues of A are positive (resp., non-negative), then A is *positive* (resp., *non-negative*) definite.

Non-negative definite operators arise frequently in the following chapters. We now study non-negative definite operators L on $H = L^2(D)$.

Lemma 1.77 For a domain D , if $G \in C(D \times D)$ is a non-negative definite function, then the integral operator L on $L^2(D)$ with kernel G is non-negative.

Proof For $u, v \in L^2(D)$, we must show

$$\langle u, Lv \rangle = \int_D \int_D u(\mathbf{x}) G(\mathbf{x}, \mathbf{y}) v(\mathbf{y}) d\mathbf{x} d\mathbf{y} \geq 0.$$

We may choose simple functions G_N of the form

$$G_N(\mathbf{x}, \mathbf{y}) = \sum_{j,k=1}^N G(\mathbf{x}_j, \mathbf{y}_k) 1_{F_j}(\mathbf{x}) 1_{F_k}(\mathbf{y}), \quad \text{for } F_j \in \mathcal{B}(D \times D)$$

and $\mathbf{x}_j, \mathbf{y}_k \in D$, such that $G_N \rightarrow G$ pointwise as $N \rightarrow \infty$ (see [Exercise 1.4](#)). Then,

$$\langle u, Lv \rangle = \lim_{N \rightarrow \infty} \sum_{j,k=1}^N \int_{F_j} u(\mathbf{x}) d\mathbf{x} G(\mathbf{x}_j, \mathbf{y}_k) \int_{F_k} v(\mathbf{y}) d\mathbf{y},$$

which is non-negative by the definition of non-negative definite function. $\square$

Lemma 1.78 (Dini) For a bounded domain D , let $f_n \in C(\bar{D})$ be such that $f_n(\mathbf{x}) \leq f_{n+1}(\mathbf{x})$ for $n \in \mathbb{N}$ and $f_n(\mathbf{x}) \rightarrow f(\mathbf{x})$ as $n \rightarrow \infty$ for all $\mathbf{x} \in \bar{D}$. Then $\|f - f_n\|_\infty \rightarrow 0$ as $n \rightarrow \infty$.

Proof Suppose for a contradiction that $\|f - f_n\|_\infty \geq \epsilon$ as $n \rightarrow \infty$ for some $\epsilon > 0$. Let $D_n := \{\mathbf{x} \in \bar{D} : f(\mathbf{x}) - f_n(\mathbf{x}) < \epsilon\}$. Then, there exists $\mathbf{x}_n \in \bar{D}$ with $\mathbf{x}_n \notin D_n$ for all n sufficiently large. Because $\bar{D}$ is compact, $\mathbf{x}_n$ has a limit point $\mathbf{x} \in \bar{D}$. As f_n is increasing, the set D_n is increasing, $D_n \subset D_{n+1}$, and $\mathbf{x} \notin D_n$ for any n . Hence, $f(\mathbf{x}) - f_n(\mathbf{x}) \geq \epsilon$ for all n . This is a contradiction as $f_n(\mathbf{x}) \rightarrow f(\mathbf{x})$. $\square$

Definition 1.79 (trace class) For a separable Hilbert space H , a non-negative definite operator $L \in \mathcal{L}(H)$ is of *trace class* if $\text{Tr } L < \infty$, where the trace is defined by $\text{Tr } L := \sum_{j=1}^\infty \langle L\phi_j, \phi_j \rangle$, for an orthonormal basis $\{\phi_j : j \in \mathbb{N}\}$. See [Exercise 1.19](#).

Theorem 1.80 (Mercer) For a bounded domain D , let $G \in C(\bar{D} \times \bar{D})$ be a symmetric and non-negative definite function and let L be the corresponding integral operator (1.17). There exist eigenfunctions ϕ_j of L with eigenvalues $\lambda_j > 0$ such that $\phi_j \in C(\bar{D})$ and

$$G(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^\infty \lambda_j \phi_j(\mathbf{x}) \phi_j(\mathbf{y}), \quad \mathbf{x}, \mathbf{y} \in \bar{D},$$

where the sum converges in $C(\bar{D} \times \bar{D})$. Furthermore,

$$\sup_{\mathbf{x}, \mathbf{y} \in \bar{D}} \left| G(\mathbf{x}, \mathbf{y}) - \sum_{j=1}^N \lambda_j \phi_j(\mathbf{x}) \phi_j(\mathbf{y}) \right| \leq \sup_{\mathbf{x} \in \bar{D}} \sum_{j=N+1}^\infty \lambda_j |\phi_j(\mathbf{x})|^2. \quad (1.22)$$

The operator L is of trace class and

$$\operatorname{Tr} L = \int_D G(\mathbf{x}, \mathbf{x}) d\mathbf{x}.$$

Proof By [Theorem 1.68](#), L is a compact linear operator on $L^2(D)$. Then, G and hence L is symmetric and [Theorem 1.73](#) applies with $H = L^2(D)$. There exists an orthonormal basis of eigenfunctions ϕ_j for the range of L . By [Lemma 1.77](#), L is non-negative definite and the eigenvalues $\lambda_j > 0$ (eigenfunctions with zero eigenvalues do not contribute to the range of L). Now $\langle \phi_j, L\phi_k \rangle_{L^2(D)} = \lambda_j$ for $j = k$ and $\langle \phi_j, L\phi_k \rangle_{L^2(D)} = 0$ for $j \neq k$, so that (1.18) gives $G(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^{\infty} \lambda_j \phi_j(\mathbf{x}) \phi_j(\mathbf{y})$. Here $\phi_j(\mathbf{x})$ is a continuous function of $\mathbf{x}$ for $\lambda_j \neq 0$, because $\phi_j(\mathbf{x}) = (1/\lambda_j) \int_D G(\mathbf{x}, \mathbf{y}) \phi_j(\mathbf{y}) d\mathbf{y}$ and the kernel G is continuous.

Define $G_N(\mathbf{x}, \mathbf{y})$ via $G_N(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^N \lambda_j \phi_j(\mathbf{x}) \phi_j(\mathbf{y})$. The function $G - G_N$ is non-negative definite, as the corresponding integral operator has non-negative eigenvalues λ_j for $j = N+1, \dots, \infty$. Under the non-negative definite condition ([Definition 1.75](#) with $a_j = 1$ and $\mathbf{x}_j = \mathbf{x}$), $G(\mathbf{x}, \mathbf{x}) - G_N(\mathbf{x}, \mathbf{x}) \geq 0$ for all $\mathbf{x} \in \bar{D}$. Let $g_N(\mathbf{x}) := G_N(\mathbf{x}, \mathbf{x})$; then g_N is increasing and bounded and has a limit, say g . The convergence is uniform by [Lemma 1.78](#) and $\|g - g_N\|_{\infty} \rightarrow 0$ as $N \rightarrow \infty$.

By the Cauchy–Schwarz inequality ([Lemma 1.32](#)),

$$\sum_{j=N+1}^{\infty} \lambda_j \phi_j(\mathbf{x}) \phi_j(\mathbf{y}) \leq \left(\sum_{j=N+1}^{\infty} \lambda_j |\phi_j(\mathbf{x})|^2 \sum_{j=N+1}^{\infty} \lambda_j |\phi_j(\mathbf{y})|^2 \right)^{1/2},$$

so that $\|G - G_N\|_{\infty} \leq \|g - g_N\|_{\infty}$, which is (1.22). Then, G is a uniform limit of G_N in $C(\bar{D} \times \bar{D})$. Finally, $G(\mathbf{x}, \mathbf{x}) = \sum_{j=1}^{\infty} \lambda_j \phi_j(\mathbf{x})^2$. Integrating over D and using the fact that $\int_D \phi_j(\mathbf{x})^2 d\mathbf{x} = 1$, we see

$$\int_D G(\mathbf{x}, \mathbf{x}) d\mathbf{x} = \sum_{j=1}^{\infty} \lambda_j = \operatorname{Tr} L.$$

As $G \in C(\bar{D} \times \bar{D})$ and D is bounded, G and the integral of G over D are bounded. Hence, $\operatorname{Tr} L < \infty$ and L has trace class. $\square$

Unbounded operators

The Hilbert–Schmidt and projection operators are examples of bounded operators. The differential operators that arise in the study of PDEs are unbounded and usually defined on a strict subset of an underlying Hilbert space H . The following example with the Laplace operator illustrates the situation well.

Example 1.81 (Laplacian with Dirichlet conditions) Consider the Dirichlet problem for the Poisson equation: for a given $f \in L^2(0, 1)$, find u such that

$$-u_{xx} = f, \quad u(0) = u(1) = 0. \quad (1.23)$$

We discuss this problem in detail in [Chapter 2](#). Here, rather than consider (1.23) with classical derivatives, we use weak derivatives and seek $u \in H^2(0, 1)$. We also assume that $u \in H_0^1(0, 1)$, the Sobolev space given in (1.47). The Sobolev embedding theorem gives

$H_0^1(0, 1) \subset C([0, 1])$ and hence the Dirichlet conditions $u(0) = u(1) = 0$ are well defined. We recast (1.23) in terms of an operator A defined by

$$Au := -u_{xx}, \quad u \in H^2(0, 1) \cap H_0^1(0, 1). \quad (1.24)$$

Note that A maps from a strict subset of $L^2(0, 1)$ to $L^2(0, 1)$ and that A is unbounded because, for $\phi_j(x) = \sqrt{2} \sin(j\pi x)$, we have

$$\|A\phi_j\|_{L^2(0,1)} = \pi^2 j^2 \|\phi_j\|_{L^2(0,1)} = \pi^2 j^2 \rightarrow \infty \quad \text{as } j \rightarrow \infty.$$

Now (1.23) becomes

$$\text{find } u \in H^2(0, 1) \cap H_0^1(0, 1) \text{ such that } Au = f. \quad (1.25)$$

Definition 1.82 (domain of linear operator) For a linear operator $A: \mathcal{D}(A) \subset H \rightarrow H$, the set $\mathcal{D}(A)$ is known as the *domain of A* .

For the operator A in [Example 1.81](#), the domain is $\mathcal{D}(A) = H^2(0, 1) \cap H_0^1(0, 1)$.

The inverse of an unbounded operator often has much nicer properties. For the operator A in (1.24), A^{-1} is symmetric and compact and the Hilbert–Schmidt spectral theorem applies. We investigate this in [Example 1.83](#). We adopt the convention that L is a bounded operator with eigenvalues ν_j and A is an unbounded operator with eigenvalues λ_j .

Example 1.83 Continuing with [Example 1.81](#) for fixed $y \in (0, 1)$, let $G(x, y)$ be the solution of

$$-G_{xx} = \delta(x - y), \quad G(0, y) = G(1, y) = 0,$$

where δ is the Dirac delta function, introduced in [Example 1.43](#). Here, G is known as the *Green's function* for the operator A . In this case (see [Exercise 1.20](#)),

$$G(x, y) = \begin{cases} x(1 - y), & x < y, \\ y(1 - x), & x \geq y. \end{cases} \quad (1.26)$$

Define

$$(Lf)(x) := \int_0^1 G(x, y)f(y) dy, \quad \forall f \in L^2(0, 1).$$

Then

$$A \int_0^1 G(x, y)f(y) dy = \int_0^1 AG(x, y)f(y) dy = \int_0^1 \delta(x - y)f(y) dy = f(x)$$

and $A(Lf) = f$, so that L is the inverse of A . The second derivatives of Lf are well defined in $L^2(0, 1)$ and it is easy to check that Lf obeys the Dirichlet boundary conditions. Hence, $Lf \in H^2(0, 1) \cap H_0^1(0, 1) = \mathcal{D}(A)$ and $u = Lf$ is the solution to (1.25).

Observe that G is a symmetric function and is square integrable. Then, the operator L is symmetric (see [Lemma 1.72](#)) and compact (see [Theorem 1.68](#)) on $L^2(0, 1)$. By [Theorem 1.73](#), there exists a decreasing set of eigenvalues ν_j corresponding to an orthonormal basis of eigenfunctions $\phi_j \in L^2(0, 1)$ of L . Now $L\phi_j = \nu_j\phi_j$. Hence, $\phi_j \in \mathcal{D}(A)$ and, applying the operator A , we have $\phi_j = \nu_j A\phi_j$. Then,

$$-A\phi_j + \frac{1}{\nu_j}\phi_j = \frac{d\phi_j^2}{dx^2} + \frac{1}{\nu_j}\phi_j = 0, \quad \phi_j(0) = 0 = \phi_j(1).$$

The solution is $v_j = 1/(\pi^2 j^2)$ for $j \in \mathbb{N}$ and the corresponding normalised eigenfunctions are $\phi_j(x) = \sqrt{2} \sin(j\pi x)$.

From (1.21),

$$Lf = \sum_{j=1}^{\infty} \frac{1}{\pi^2 j^2} \langle f, \phi_j \rangle_{L^2(0,1)} \phi_j, \quad \forall f \in L^2(0,1).$$

Because $\{\phi_j\}$ is an orthonormal basis for $L^2(0,1)$ and A is symmetric,

$$Au = \sum_{j=1}^{\infty} \langle Au, \phi_j \rangle_{L^2(0,1)} \phi_j = \sum_{j=1}^{\infty} \langle u, A\phi_j \rangle_{L^2(0,1)} \phi_j$$

and, since $A\phi_j = \pi^2 j^2 \phi_j$, $\lambda_j = \pi^2 j^2$ are the eigenvalues of A and

$$Au = \sum_{j=1}^{\infty} \pi^2 j^2 \langle u, \phi_j \rangle_{L^2(0,1)} \phi_j, \quad \forall u \in \mathcal{D}(A).$$

Hence, we have derived a representation for A in terms of the eigenfunctions ϕ_j .

Example 1.84 (periodic boundary conditions) The Laplace operator with periodic boundary conditions maps constant functions to zero and does not have a well-defined inverse or Green's function. Instead, we consider the following boundary-value problem: find u such that

$$u - u_{xx} = f, \quad u(-\pi) = u(\pi), \quad u_x(-\pi) = u_x(\pi), \quad (1.27)$$

for a given $f \in L^2(-\pi, \pi)$. Here, we choose the space $H_{\text{per}}^2(-\pi, \pi)$ for the domain. Again, $H_{\text{per}}^2(-\pi, \pi) \subset C^1([-\pi, \pi])$ by the Sobolev embedding theorem and the boundary conditions are well defined. Define the operator A by

$$Au := u - u_{xx} \quad \text{for } u \in \mathcal{D}(A) = H_{\text{per}}^2(-\pi, \pi),$$

so that (1.27) becomes: find $u \in \mathcal{D}(A)$ such that $Au = f$. The Green's function for this problem is the solution of $G - G_{xx} = \delta(x - y)$, subject to the periodic boundary conditions, and is a symmetric and square integrable function (see [Exercise 1.20](#)).

Once again, the Hilbert–Schmidt spectral theory applies to $L = A^{-1}$ and the eigenvalues v of L are found by solving

$$-A\phi + \frac{1}{v}\phi = -\phi + \frac{d\phi^2}{dx^2} + \frac{1}{v}\phi = 0,$$

subject to $\phi(-\pi) = \phi(\pi)$ and $\phi_x(-\pi) = \phi_x(\pi)$. Using complex notation, the eigenvalues of L are $v_k = 1/(1 + k^2)$ for $k \in \mathbb{Z}$ and the corresponding eigenfunctions are $\psi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}$. Therefore, A has eigenvalues $\lambda_k = 1/v_k = 1 + k^2$ and, for $u \in \mathcal{D}(A)$,

$$Au = \sum_{k \in \mathbb{Z}} (1 + k^2) u_k \psi_k, \quad u_k = \langle u, \psi_k \rangle_{L^2((-\pi, \pi), \mathbb{C})}.$$

When u is a real-valued function, $u_{-k} = \bar{u}_k$ and Au is also real valued.

In [Theorem 1.73](#), we index the eigenvalues λ_j of L by $j \in \mathbb{N}$ and the corresponding

eigenfunctions $\phi_j \in L^2(-\pi, \pi)$ and are real-valued functions. Here, we still have the representation

$$Au = \sum_{j=1}^{\infty} \lambda_j u_j \phi_j, \quad u_j = \langle u, \phi_j \rangle_{L^2(-\pi, \pi)}, \quad u \in \mathcal{D}(A)$$

if, we set for $k \in \mathbb{N}$,

$$\begin{aligned} \phi_1(x) &= \frac{1}{\sqrt{2\pi}}, & \phi_{2k}(x) &= \frac{1}{\sqrt{\pi}} \sin(kx), & \phi_{2k+1}(x) &= \frac{1}{\sqrt{\pi}} \cos(kx), \\ \lambda_1 &= 1, & \lambda_{2k} &= 1 + k^2, & \lambda_{2k+1} &= 1 + k^2. \end{aligned} \quad (1.28)$$

Powers of a linear operator

Fractional powers, A^α for $\alpha \in \mathbb{R}$, of a linear operator A are useful for understanding Sobolev spaces and analysing PDEs. Here, we take a simple approach and assume the following.

Assumption 1.85 A is a linear operator from $\mathcal{D}(A) \subset H$ to a Hilbert space H with an orthonormal basis of eigenfunctions $\{\phi_j : j \in \mathbb{N}\}$ and eigenvalues $\lambda_j > 0$ ordered so that $\lambda_{j+1} \geq \lambda_j$.

In [Examples 1.83](#) and [1.84](#), we have already met operators satisfying [Assumption 1.85](#) with $H = L^2(D)$. All operators of this type are self-adjoint.

Lemma 1.86 (self-adjoint) *Let A obey Assumption 1.85. Then, $\langle Au, v \rangle = \langle u, Av \rangle$ for all $u, v \in \mathcal{D}(A)$. We say A is self-adjoint.*

Proof If the assumption holds then $Au = \sum_{j=1}^{\infty} \lambda_j \langle u, \phi_j \rangle \phi_j$ for $u \in \mathcal{D}(A)$. Hence,

$$\langle Au, v \rangle = \sum_{j=1}^{\infty} \lambda_j \langle u, \phi_j \rangle \langle v, \phi_j \rangle = \langle u, Av \rangle. \quad \square$$

As the eigenvalues are strictly positive, λ_j^α is well defined in $\mathbb{R}^+$ for any $\alpha \in \mathbb{R}$ and we have the following definition.

Definition 1.87 (fractional powers of a linear operator) Under [Assumption 1.85](#), define the fractional power $A^\alpha u$ for $\alpha \in \mathbb{R}$ by

$$A^\alpha u := \sum_{j=1}^{\infty} \lambda_j^\alpha u_j \phi_j, \quad \text{for } u = \sum_{j=1}^{\infty} u_j \phi_j, \quad u_j \in \mathbb{R},$$

and let the domain $\mathcal{D}(A^\alpha)$ be the set of $u = \sum_{j=1}^{\infty} u_j \phi_j$ such that $A^\alpha u \in H$.

Theorem 1.88 *Under Assumption 1.85, the domain $\mathcal{D}(A^\alpha)$ for $\alpha \in \mathbb{R}$ is a Hilbert space with inner product $\langle u, v \rangle_\alpha := \langle A^\alpha u, A^\alpha v \rangle$ and corresponding induced norm $\|u\|_\alpha := \|A^\alpha u\|$.*

We list some elementary properties of $\mathcal{D}(A^\alpha)$.

Lemma 1.89 *Let Assumption 1.85 hold.*

- (i) $\|u\|_0 = \|u\|$ for $u \in \mathcal{D}(A^0) = H$.
- (ii) $\langle u, v \rangle_{1/2} = \langle A^{1/2}u, A^{1/2}v \rangle = \langle Au, v \rangle$ for $u \in \mathcal{D}(A)$ and $v \in \mathcal{D}(A^{1/2})$.
 $\langle u, v \rangle = \langle A^{1/2}u, A^{-1/2}v \rangle$ for $u \in \mathcal{D}(A^{1/2})$ and $v \in H$.
- (iii) For $\alpha > 0$, $\|u\|_\alpha \geq \lambda_1^\alpha \|u\|$ for $u \in \mathcal{D}(A^\alpha) \subset H$.
- (iv) For $\alpha < 0$, $H \subset \mathcal{D}(A^\alpha)$ and $A^\alpha \in \mathcal{L}(H)$ with $\|A^\alpha\|_{\mathcal{L}(H)} = \lambda_1^\alpha$.

Proof The result is an elementary consequence of Definition 1.87 and the fact that λ_j is a non-decreasing sequence. For example, for $\alpha < 0$, A^α has non-increasing eigenvalues λ_j^α bounded from above by λ_1^α and

$$\|A^\alpha u\| = \left(\sum_{j=1}^{\infty} \lambda_j^{2\alpha} |u_j|^2 \right)^{1/2} \leq \lambda_1^\alpha \|u\|$$

with equality if $u = \phi_1$. Then, $\|A^\alpha\|_{\mathcal{L}(H)} = \sup_{u \neq 0} \|A^\alpha u\| / \|u\| = \lambda_1^\alpha$. $\square$

Example 1.90 (fractional powers for Laplacian with Dirichlet conditions) Continuing with [Example 1.83](#), the Laplace operator $Au = -u_{xx}$ with domain $\mathcal{D}(A) = H^2(0, 1) \cap H_0^1(0, 1)$ has eigenfunctions $\phi_j(x) = \sqrt{2} \sin(j\pi x)$ and eigenvalues $\lambda_j = j^2\pi^2$ for $j \in \mathbb{N}$. For $u = \sum_j u_j \phi_j \in \mathcal{D}(A^\alpha)$, the fractional power A^α and fractional power norm $\|\cdot\|_\alpha$ are then given by

$$A^\alpha u = \sum_{j=1}^{\infty} (j\pi)^{2\alpha} u_j \phi_j, \quad \|u\|_\alpha = \left(\sum_{j=1}^{\infty} (j\pi)^{4\alpha} u_j^2 \right)^{1/2}.$$

Example 1.91 (fractional power with periodic conditions) Consider the operator $Au = u - u_{xx}$ with domain $\mathcal{D}(A) = H_{\text{per}}^2(-\pi, \pi)$ from [Example 1.84](#). We write the definition of A^α in complex notation using the eigenfunctions $\psi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}$ and eigenvalues $\lambda_k = 1 + k^2$ indexed by $k \in \mathbb{Z}$. Writing $u = \sum_{k \in \mathbb{Z}} u_k \psi_k$, we have

$$A^\alpha u = \sum_{k \in \mathbb{Z}} (1 + k^2)^\alpha u_k \psi_k, \quad \|u\|_\alpha = \left(\sum_{k \in \mathbb{Z}} (1 + k^2)^{2\alpha} |u_k|^2 \right)^{1/2}.$$

Once again, when $u_{-k} = \bar{u}_k$, both u and $A^\alpha u$ are real valued.

Example 1.92 (Dirac delta function) Continuing from [Example 1.91](#), we show that the Dirac delta function belongs to $\mathcal{D}(A^\alpha)$ for $\alpha < -1/4$ and that $\mathcal{D}(A^\alpha)$ is strictly larger than $L^2(-\pi, \pi)$. Although the delta function is not a function, formally we can write

$$\delta = \sum_{k \in \mathbb{Z}} \langle \delta, \psi_k \rangle_{L^2((-\pi, \pi), \mathbb{C})} \psi_k = \sum_{k \in \mathbb{Z}} \overline{\psi_k(0)} \psi_k \quad (1.29)$$

because, for continuous functions ϕ ,

$$\langle \delta, \phi \rangle_{L^2((-\pi, \pi), \mathbb{C})} = \int_{-\pi}^{\pi} \delta(x) \overline{\phi(x)} dx = \overline{\phi(0)}.$$

Substituting for ψ_k , $\delta(x) = \frac{1}{2\pi} \sum_{k \in \mathbb{Z}} e^{ikx}$ and

$$A^\alpha \delta(x) = \frac{1}{2\pi} \sum_{k \in \mathbb{Z}} (1 + k^2)^\alpha e^{ikx}.$$

The sum is real valued as the imaginary parts of $e^{\pm ikx}$ cancel. Although δ and the series (1.29) are not well defined in $L^2(-\pi, \pi)$, the fractional power $A^\alpha \delta$ is well defined in $L^2(-\pi, \pi)$ if $\alpha < -1/4$, as

$$\|A^\alpha \delta\|_{L^2(-\pi, \pi)} = \left(\frac{1}{2\pi} \sum_{k \in \mathbb{Z}} (1 + k^2)^{2\alpha} \right)^{1/2}.$$

We conclude that the delta function $\delta \in \mathcal{D}(A^\alpha)$ for $\alpha < -1/4$.

Fractional powers have a number of important applications. In §3.3, they are used to analyse the exponential e^{-At} of a linear operator. We now give two applications: the first shows a relationship between the Sobolev spaces $H^r(-\pi, \pi)$ and fractional powers of the Laplacian. The second describes the approximation of u by its projection onto the first N eigenfunctions.

Proposition 1.93 *For $D = (-\pi, \pi)$, let $Au = u - u_{xx}$ with domain $\mathcal{D}(A) = H_{\text{per}}^2(-\pi, \pi)$ and let $\|\cdot\|_\alpha$ denote the corresponding fractional power norm. For $r \in \mathbb{N}$, the $H^r(-\pi, \pi)$ norm (see [Definition 1.44](#)) is equivalent to the fractional power norm $\|u\|_{r/2}$.*

Proof For $u \in L^2(-\pi, \pi)$, write

$$u(x) = \frac{1}{\sqrt{2\pi}} \sum_{k \in \mathbb{Z}} u_k e^{ikx}$$

for some $u_k \in \mathbb{C}$ with $u_{-k} = \bar{u}_k$. Differentiation of u with respect to x results in the coefficient u_k being multiplied by ik . In terms of the Sobolev norms, this means

$$\|u\|_{H^r(-\pi, \pi)}^2 = \sum_{0 \leq |\alpha| \leq r} \|\mathcal{D}^\alpha u\|_{L^2(-\pi, \pi)}^2 = \sum_{k \in \mathbb{Z}} (1 + k^2 + \dots + k^{2r}) |u_k|^2. \quad (1.30)$$

The condition $u \in H^r(-\pi, \pi)$ now concerns the decay of the coefficients $|u_k|$ as $|k| \rightarrow \infty$. Note that $1 + k^2 \leq 2k^2$ for $k \neq 0$ and

$$\frac{1}{2^r} (1 + k^2)^r \leq (1 + k^2 + \dots + k^{2r}) \leq (1 + k^2)^r.$$

As $\|u\|_{r/2}^2 = \sum_{k \in \mathbb{Z}} (1 + k^2)^r |u_k|^2$, we obtain $\frac{1}{2^r} \|u\|_{r/2}^2 \leq \|u\|_{H^r(-\pi, \pi)}^2 \leq \|u\|_{r/2}^2$, which gives the required norm equivalence. $\square$

We have treated, for simplicity, the case $r \in \mathbb{N}$, but the proposition can be extended when $H^r(-\pi, \pi)$ is defined for negative and non-integer values r . In that case, following [Example 1.92](#), we can say the delta function $\delta \in H^r(-\pi, \pi)$ for $r < -1/2$.

Finally, we study the orthogonal projection P_N from H onto the first N eigenfunctions of A (see [Lemma 1.41](#)). $P_N u$ is often used to approximate u and, when $u \in \mathcal{D}(A^\alpha)$ for some $\alpha > 0$, we have the following estimate on the approximation error $\|P_N u - u\|$. We see smoother functions (where α is large) are easier to approximate.

Lemma 1.94 *Let Assumption 1.85 hold. For $\alpha > 0$, $\|u - P_N u\| \leq \lambda_{N+1}^{-\alpha} \|u\|_\alpha$ for $u \in \mathcal{D}(A^\alpha)$ and $\|A^{-\alpha}(I - P_N)\|_{\mathcal{L}(H)} = \lambda_{N+1}^{-\alpha}$.*

Proof With $u_j = \langle u, \phi_j \rangle$, we have

$$u - P_N u = \sum_{j=N+1}^{\infty} u_j \phi_j = \sum_{j=N+1}^{\infty} \lambda_j^{-\alpha} (\lambda_j^{\alpha} u_j) \phi_j.$$

As $\lambda_j^{-\alpha}$ is non-increasing, $\|u - P_N u\| \leq \lambda_{N+1}^{-\alpha} \|u\|_{\alpha}$. Further,

$$\|A^{-\alpha}(I - P_N)u\| = \left(\sum_{j=N+1}^{\infty} \lambda_j^{-2\alpha} |u_j|^2 \right)^{1/2} \leq \lambda_{N+1}^{-\alpha} \|(I - P_N)u\| \leq \lambda_{N+1}^{-\alpha} \|u\|,$$

with equality if $u = \phi_{N+1}$. The result now follows from the definition of the operator norm (Lemma 1.54). $\square$

1.4 Fourier analysis

In Fourier analysis, a function is decomposed into a linear combination of elementary waves of different frequencies. These waves are typically sines or cosines (often written using the complex exponential) and coincide with the eigenfunctions of the Laplacian (see Examples 1.83 and 1.84). A given function u in a Hilbert space (the so-called physical space) can be written as a linear combination of waves, and the set of coefficients in this expansion belong to a second Hilbert space, known as the frequency domain or Fourier space. The mapping between the physical space and the frequency domain is known as a Fourier transform, and it is an invertible mapping that preserves the inner products and hence the norms on the Hilbert spaces up to a scalar constant. In other words, the Fourier transform is an isometry. Many problems admit elegant solutions in the frequency domain, as we see when studying differential equations and stochastic processes in later chapters. We briefly review the following four tools from Fourier analysis:

The discrete Fourier transform (DFT) which maps vectors in $\mathbb{C}^d$ to vectors in $\mathbb{C}^d$.

Fourier series which maps functions $u \in L^2((a, b), \mathbb{C})$ to a set of coefficients $\{u_k\} \in \ell^2 := L^2(\mathbb{Z}, \mathbb{C})$.

The Fourier transform which maps functions in $L^2(\mathbb{R}^d, \mathbb{C})$ to functions in $L^2(\mathbb{R}^d, \mathbb{C})$.

The Hankel transform which maps isotropic functions in $L^2(\mathbb{R}^d, \mathbb{C})$ to isotropic functions in $L^2(\mathbb{R}^d, \mathbb{C})$.

Discrete Fourier Transform

The discrete Fourier transform (DFT) is a key computational tool for applying Fourier transforms. For example, it is used in Chapter 3 for the numerical solution of PDEs and in Chapters 6 and 7 to generate approximations of Gaussian random fields. The DFT gives an expression for a vector $\mathbf{u} \in \mathbb{C}^d$ in terms of the columns of the following matrix.

Definition 1.95 (Fourier matrix) The Fourier matrix W is the $d \times d$ matrix with complex entries $w_{\ell m} = \omega^{(\ell-1)(m-1)} / \sqrt{d}$, where $\omega := e^{-2\pi i/d}$ is the d th root of unity.

The Fourier matrix is unitary (i.e., $W^*W = I$ where $*$ denotes conjugate transpose), as is verified by the following calculation. Let the m th column of W be $\mathbf{w}_m$. Then,

$$\mathbf{w}_i^* \mathbf{w}_m = \frac{1}{d} \sum_{\ell=1}^d \omega^{-(\ell-1)(i-1)} \omega^{(\ell-1)(m-1)} = \frac{1}{d} \sum_{\ell=1}^d \omega^{(\ell-1)(m-i)} = \delta_{im},$$

where δ_{im} is the Kronecker delta function. In other words, the vectors $\{\mathbf{w}_m : m = 1, \dots, d\}$ form an orthonormal basis for $\mathbb{C}^d$, which is known as the Fourier basis. It is the expression of vectors in $\mathbb{C}^d$ in the Fourier basis that is of interest.

Definition 1.96 (discrete Fourier transform) The DFT of $\mathbf{u} \in \mathbb{C}^d$ is the vector $\hat{\mathbf{u}} = [\hat{u}_1, \dots, \hat{u}_d]^T \in \mathbb{C}^d$ given by

$$\hat{u}_\ell := \sum_{m=1}^d u_m \omega^{(\ell-1)(m-1)}, \quad \ell = 1, \dots, d, \quad \omega := e^{-2\pi i/d}. \quad (1.31)$$

In many applications, $\mathbf{u} \in \mathbb{R}^d$ (is real valued), in which case $\hat{u}_\ell = \bar{\hat{u}}_{d-\ell}$. The *inverse discrete Fourier transform (IDFT)* of $\hat{\mathbf{u}} \in \mathbb{C}^d$ is the vector $\mathbf{u} \in \mathbb{C}^d$ defined by

$$u_\ell := \frac{1}{d} \sum_{m=1}^d \hat{u}_m \omega^{-(m-1)(\ell-1)}, \quad \ell = 1, \dots, d. \quad (1.32)$$

The coefficients $\hat{u}_\ell$ of the DFT are *not* the coefficients of $\mathbf{u}$ in the Fourier basis $\mathbf{w}_m$ and in fact $\hat{\mathbf{u}} = \sqrt{d} W \mathbf{u}$. The choice of scaling means that

$$\|\hat{\mathbf{u}}\|_2 = \sqrt{d} \|\mathbf{u}\|_2, \quad \hat{\mathbf{u}}^T \hat{\mathbf{v}} = d \mathbf{u}^T \mathbf{v}, \quad (1.33)$$

which are known as the Plancherel and Parseval identities respectively. This scaling is used in computing environments such as MATLAB and (1.31)–(1.32) agree with the MATLAB commands `fft` and `ifft` respectively. Routines such as `fft` implement the fast Fourier transform (FFT) algorithm and are very efficient, especially when d is a power of 2, running in $\mathcal{O}(d \log d)$ operations. Compare this to a standard matrix vector multiplication, which requires $\mathcal{O}(d^2)$ operations.

Fourier series $L^2((a, b), \mathbb{C}) \longleftrightarrow L^2(\mathbb{Z}, \mathbb{C})$

As in [Example 1.84](#), the functions $\frac{1}{\sqrt{2\pi}} e^{ikx}$ form an orthonormal basis for $L^2((-\pi, \pi), \mathbb{C})$ and, similarly, $\psi_k(x) := \frac{1}{\sqrt{b-a}} e^{2\pi i k x / (b-a)}$ form an orthonormal basis for $L^2((a, b), \mathbb{C})$. Hence, any $u \in L^2((a, b), \mathbb{C})$ can be written as

$$u(x) = \frac{1}{\sqrt{b-a}} \sum_{k=-\infty}^{\infty} u_k e^{2\pi i k x / (b-a)}, \quad u_k = \langle u, \psi_k \rangle_{L^2((a, b), \mathbb{C})}.$$

In the real-valued case, $u \in L^2(a, b)$ and $u_{-k} = \bar{u}_k$. This expansion is called a Fourier series and provides a mapping from the space $L^2((a, b), \mathbb{C})$ to the set of Fourier coefficients $\{u_k\} \in L^2(\mathbb{Z}, \mathbb{C})$. Here, $L^2(\mathbb{Z}, \mathbb{C})$ is the Hilbert space of sequences $\{u_k : k \in \mathbb{Z}\}$ that are square integrable, which we denote ℓ^2 , and has inner product $\langle \{u_k\}, \{v_k\} \rangle_{\ell^2} := \sum_k u_k \bar{v}_k$.

It is convenient to rescale the above equations when defining the Fourier series.

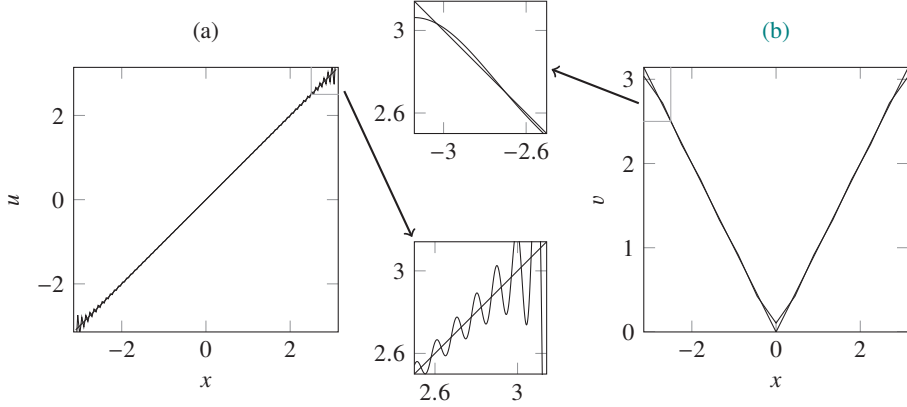


Figure 1.4 Plots of (a) $u(x) = x$ and the series (1.37) truncated to 64 terms and (b) $v(x) = |x|$ and the series (1.38) truncated to 8 terms, with an enlargement of the behaviour near to the boundary. The large oscillations seen in (a) are called the Gibbs phenomenon and occur because $u(-\pi) \neq u(\pi)$.

Definition 1.97 (Fourier series) Let $u \in L^2((a, b), \mathbb{C})$. The Fourier series is

$$u(x) = \sum_{k=-\infty}^{\infty} u_k e^{2\pi i k x / (b-a)}, \quad (1.34)$$

where the Fourier coefficients $\{u_k\} \in \ell^2$ are defined by

$$u_k := \frac{1}{b-a} \int_a^b u(x) e^{-2\pi i k x / (b-a)} dx. \quad (1.35)$$

With this scaling, the Parseval identity says that

$$\langle u, v \rangle_{L^2((a,b), \mathbb{C})} = (b-a) \langle \{u_k\}, \{v_k\} \rangle_{\ell^2}. \quad (1.36)$$

Example 1.98 (explicit calculations of coefficients) The function $u(x) = x$ on $(-\pi, \pi)$ has Fourier coefficients $u_0 = 0$ and

$$u_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-ikx} dx = -\frac{i}{2\pi} \int_{-\pi}^{\pi} x \sin(kx) dx = \frac{i}{k} (-1)^k, \quad \text{for } k \neq 0.$$

The function $v(x) = |x|$ on $(-\pi, \pi)$ has Fourier coefficients $v_0 = \pi/2$ and

$$v_k = \frac{1}{\pi} \int_0^{\pi} x e^{-ikx} dx = \frac{1}{\pi} \int_0^{\pi} x \cos(kx) dx = \frac{1}{\pi k^2} (\cos(k\pi) - 1), \quad \text{for } k \neq 0.$$

The coefficients v_k are real because $v(x)$ is an even function. We also have

$$u(x) = \sum_{k=-\infty}^{\infty} u_k e^{ikx} = \sum_{j=1}^{\infty} \frac{2}{j} (-1)^{j+1} \sin(jx), \quad (1.37)$$

$$v(x) = \sum_{k=-\infty}^{\infty} v_k e^{ikx} = \frac{\pi}{2} + \sum_{j=1}^{\infty} \frac{2(\cos(j\pi) - 1)}{\pi j^2} \cos(jx). \quad (1.38)$$

Odd functions can always be written as Fourier sine series and even functions can be written as Fourier cosine series. See Figure 1.4 and Exercise 1.28. Notice that $|u_k| = \mathcal{O}(1/k)$ and

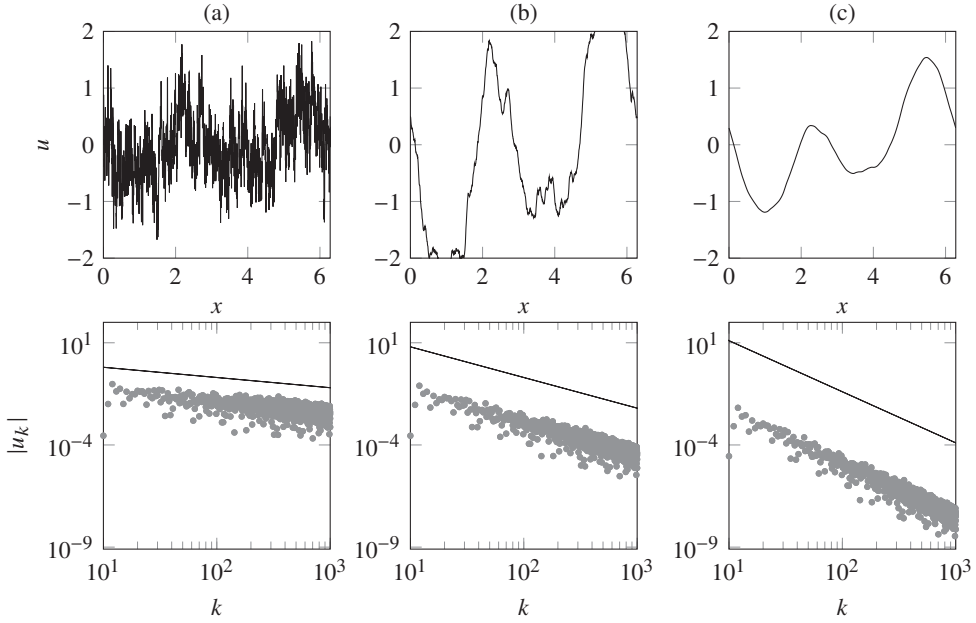


Figure 1.5 For three sets of Fourier coefficients u_k , the lower plots show $|u_k|$ against k and a line with slope (a) $-1/2$, (b) $-3/2$, (c) $-5/2$ showing the rate of decay of u_k . The upper plots show the corresponding function $u(x)$ on $[0, 2\pi]$ given by (1.34) and we know that (a) $u \in L^2(0, 2\pi)$, (b) $u \in H^1(0, 2\pi)$, and (c) $u \in H^2(0, 2\pi)$. The function $u(x)$ is smoother when u_k decays more rapidly.

$|v_k| = \mathcal{O}(1/k^2)$. The rapid rate of decay of $|v_k|$ compared to $|u_k|$ results from fact that the periodic extension of v is continuous while that of u is discontinuous. The rate of decay of Fourier coefficients is of keen interest in later chapters.

Example 1.99 (smoothness) To illustrate how the rate of decay of u_k determines the smoothness of the function $u(x)$, Figure 1.5 gives three sets of Fourier coefficients u_k with rates of decay (a) $-1/2$, (b) $-3/2$, and (c) $-5/2$. The corresponding functions $u(x)$, defined by (1.34) on $[0, 2\pi]$, are shown in the upper plots of Figure 1.5 and belong to (a) $L^2(0, 2\pi)$, (b) $H^1(0, 2\pi)$, and (c) $H^2(0, 2\pi)$ respectively. Visually, we see a change in the smoothness of $u(x)$ as the decay rate of u_k is varied and in fact $u \in H^r(0, 2\pi)$ if $u_k = \mathcal{O}(|k|^{-(2r+1+\epsilon)/2})$ for some $\epsilon > 0$ (see Exercise 1.25).

In general, to use the Fourier series of a function $u(x)$, we find the coefficients u_k via a numerical approximation and truncate the Fourier series (1.34) to a finite number of terms. We now describe how to approximate u_k . For an even integer J , consider the set of points $x_j = a + jh$ for $h = (b - a)/J$ and $j = 0, \dots, J$. We approximate u_k by U_k , the composite trapezium rule approximation (see (A.1)) to the integral (1.35). That is,

$$U_k := \frac{h}{b-a} \left(\frac{u(a)e^{-2\pi i k a/(b-a)}}{2} + \sum_{j=1}^{J-1} u(x_j) e^{-2\pi i k x_j/(b-a)} + \frac{u(b)e^{-2\pi i k b/(b-a)}}{2} \right). \quad (1.39)$$

Given U_k , we define $I_J u(x)$ by

$$I_J u(x) := \sum_{k=-J/2+1}^{J/2} U_k e^{2\pi i k x / (b-a)} \quad (1.40)$$

and use it to approximate $u(x)$. To evaluate U_k efficiently, we write

$$U_k = \frac{h}{b-a} e^{-2\pi i k a / (b-a)} \left(\frac{u(a) + u(b)}{2} + \sum_{j=1}^{J-1} u(x_j) e^{-2\pi i j k / J} \right). \quad (1.41)$$

A DFT (1.31) of the vector $[(u(a) + u(b))/2, u(x_1), \dots, u(x_{J-1})]^T \in \mathbb{R}^J$ gives the bracketed term and it is efficiently evaluated for $k = 0, \dots, J-1$ by applying the FFT. Using the symmetry $u_{-k} = \bar{u}_k$, we compute the required values of U_k with [Algorithm 1.1](#).

If $u \in H_{\text{per}}^2(a, b)$, $I_J u$ is a good approximation to u in the following sense.

Lemma 1.100 *With $I_J u$ defined by (1.40), there exists $K > 0$ such that*

$$\|u - I_J u\|_{H^1(a,b)} \leq \frac{K}{J^{1/2}} \|u\|_{H^2(a,b)}, \quad \forall u \in H_{\text{per}}^2(a,b).$$

Proof See [Exercise 1.23](#). □

Example 1.101 (estimating a Sobolev norm) As we know all the coefficients of $I_J u$ in Fourier space, $\|I_J u\|_{L^2(a,b)}$ and $\|I_J u\|_{H^1(a,b)}$ are easy to evaluate from U_k and provide estimates of $\|u\|_{H^1(a,b)}$ and $\|u\|_{H^2(a,b)}$. In particular, from (1.14) and (1.36),

$$\begin{aligned} \|u\|_{L^2(a,b)} &\approx \|I_J u\|_{L^2(a,b)} = \left((b-a) \sum_{k=-J/2+1}^{J/2} |U_k|^2 \right)^{1/2}, \\ \|u\|_{H^1(a,b)} &\approx \|I_J u\|_{H^1(a,b)} = \left(\|u\|_{L^2(a,b)}^2 + (b-a) \sum_{k=-J/2+1}^{J/2} \frac{(2\pi k)^2}{(b-a)^2} |U_k|^2 \right)^{1/2}. \end{aligned}$$

See [Algorithm 1.2](#). If $u \in H_{\text{per}}^2(a, b)$, these approximations converge with $\mathcal{O}(J^{-2})$ (see [Exercise 1.24](#)). In [Figure 1.6](#), we show the error in approximating the norms of the functions $u_1(x) = x^2(x-1)^2$ and $u_2(x) = \cos(\pi x)$ on $[0, 1]$. For $u_1(x)$, the correct answer is computed to machine accuracy with $J \approx 10^3$. However, because $u_2(0) \neq u_2(1)$, the approximation to $\|u_2\|_{H^1(0,1)}$ diverges.

Algorithm 1.1 Code to find U_k as defined by (1.39) for $k = -J/2 + 1, \dots, J/2$. The inputs are a vector `u` of values $u(x)$ on the grid $x_j = a + jh$ for $j = 0, \dots, J$ and $h = (b-a)/J$; the grid spacing `h`; and `a, b` to define the interval (a, b) . The outputs are a vector `Uk` of U_k and a vector `nu` of the corresponding frequencies $2\pi k/(b-a)$.

```

1 function [Uk, nu] = get_coeffs(u, a, b)
2 J=length(u)-1; h=(b-a)/J; u1=[(u(1)+u(J+1))/2; u(2:J)];
3 Uk=(h/(b-a))*exp(-2*pi*sqrt(-1)*[0:J-1]'*a/(b-a)).*fft(u1);
4 assert(mod(J,2)==0); % J must be even
5 nu=2*pi/(b-a)*[0:J/2,-J/2+1:-1]';
```

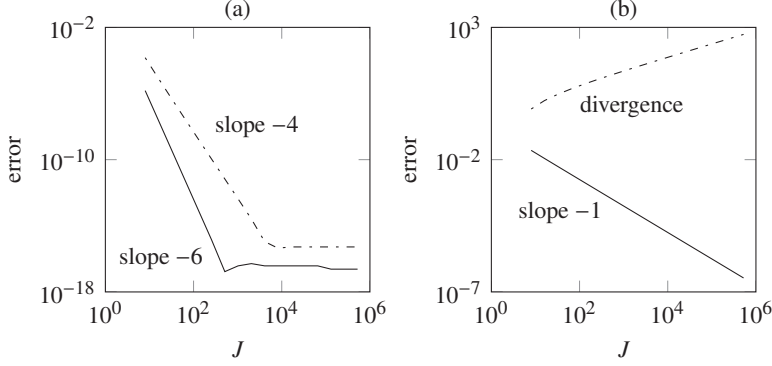


Figure 1.6 The error in approximating $\|u\|$ with $\|I_J u\|$ for the $L^2(0,1)$ (solid) and $H^1(0,1)$ (dash-dot) norm of (a) $u_1(x) = x^2(x-1)^2$ and (b) $u_2(x) = \cos(x\pi/2)$ against the number of points J . We see divergence of the approximation to the $H^1(0,1)$ norm in (b), because $u_2(x)$ is not in $H^2_{\text{per}}(0,1)$.

Algorithm 1.2 Code to approximate the $L^2(a,b)$ and $H^1(a,b)$. The inputs are a function handle `fhandle` to evaluate u , the domain `a, b`, and discretisation parameter `J`. The outputs are the $L^2(a,b)$ and $H^1(a,b)$ norms of $I_J u$.

```

1 function [l2_norm, h1_norm] = get_norm(fhandle, a, b, J)
2 grid=[a:(b-a)/J:b]';
3 % evaluate the function on the grid
4 u=fhandle(grid);
5 [Uk,nu]=get_coeffs(u, a, b);
6 l2_norm=sqrt(b-a)* norm(Uk);
7 dUk=nu.*Uk;
8 h1_norm = norm([l2_norm, sqrt(b-a)*norm(dUk)]);

```

Fourier transform $L^2(\mathbb{R}^d, \mathbb{C}) \longleftrightarrow L^2(\mathbb{R}^d, \mathbb{C})$

The Fourier series on an interval (a,b) arises from the orthonormal basis provided by the eigenfunctions of the Laplacian with periodic boundary conditions. On the whole of $\mathbb{R}^d$, $e^{i\lambda^T x}$ is an eigenfunction of the Laplacian for every $\lambda \in \mathbb{C}^d$ and, to include every wave number λ in the Fourier representation of $u \in L^2(\mathbb{R}^d, \mathbb{C})$, integrals are used.

Definition 1.102 (Fourier transform) For $u \in L^2(\mathbb{R}^d, \mathbb{C})$, the Fourier transform

$$\hat{u}(\lambda) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} u(x) e^{-i\lambda^T x} dx, \quad \lambda \in \mathbb{R}^d.$$

The inverse mapping $\hat{u} \mapsto u$ is denoted $\check{u}$ and defined via

$$\check{u}(x) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} u(\lambda) e^{i\lambda^T x} d\lambda. \quad (1.42)$$

The Parseval identity says that, for $u, v \in L^2(\mathbb{R}^d, \mathbb{C})$,

$$\langle u, v \rangle_{L^2(\mathbb{R}^d, \mathbb{C})} = \langle \hat{u}, \hat{v} \rangle_{L^2(\mathbb{R}^d, \mathbb{C})}, \quad \|u\|_{L^2(\mathbb{R}^d, \mathbb{C})} = \|\hat{u}\|_{L^2(\mathbb{R}^d, \mathbb{C})}. \quad (1.43)$$

Further properties of the Fourier transform are considered in Exercise 1.21.

Example 1.103 (Fourier transform of an exponential function) Let $d = 1$ and consider $u(x) = e^{-a|x|}$ for some $a > 0$. The Fourier transform is

$$\hat{u}(\lambda) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2 + \lambda^2}, \quad (1.44)$$

as is easily verified by integrating or using a computer algebra system.

Example 1.104 (Fourier transform of Gaussian functions) For $d = 1$ and $a > 0$, the Gaussian function $u(x) = e^{-ax^2}$ has Fourier transform

$$\hat{u}(\lambda) = \frac{1}{\sqrt{2a}} e^{-\lambda^2/4a}. \quad (1.45)$$

For a $d \times d$ symmetric positive-definite matrix C , the function $u(\mathbf{x}) = e^{-\mathbf{x}^\top C \mathbf{x}}$ has Fourier transform

$$\hat{u}(\lambda) = \frac{1}{2^{d/2} \sqrt{\det C}} e^{-\lambda^\top C^{-1} \lambda/4}. \quad (1.46)$$

Example 1.105 (Fourier transform of derivatives) For $u \in L^2(\mathbb{R}, \mathbb{C})$, the Fourier transform of the derivative $\frac{du}{dx} = \mathcal{D}u$ is found by integrating by parts,

$$\widehat{\mathcal{D}u}(\lambda) = \frac{1}{(2\pi)^{1/2}} \int_{\mathbb{R}} e^{-i\lambda x} \mathcal{D}u(x) dx = -\frac{1}{(2\pi)^{1/2}} \int_{\mathbb{R}} \frac{d}{dx} e^{-i\lambda x} u(x) dx = (i\lambda) \hat{u}(\lambda),$$

where the boundary term disappears since $|u(x)| \rightarrow 0$ as $|x| \rightarrow \infty$. We see that the derivative of the Fourier transform equals $i\lambda$ times the Fourier transform $\hat{u}(\lambda)$.

Fourier integrals for isotropic functions in $L^2(\mathbb{R}^d, \mathbb{C})$

Definition 1.106 (isotropic) A function $u: \mathbb{R}^d \rightarrow \mathbb{C}$ is *isotropic* if $u(\mathbf{x})$ depends only on $r := \|\mathbf{x}\|_2$.

The Fourier transform of an isotropic function has a special form, known as the Hankel transform, which is used in [Chapter 7](#) to develop the turning bands algorithm. To derive this special form, we make use of the Bessel function of the first kind $J_p(r)$ of order p . For $p = -1/2, 0, 1/2$, we have

$$J_{-1/2}(r) = \sqrt{\frac{2}{\pi r}} \cos(r), \quad J_0(r) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ir \cos \theta} d\theta, \quad J_{1/2}(r) = \sqrt{\frac{2}{\pi r}} \sin(r), \quad (1.47)$$

and these functions feature in the Fourier transform of isotropic functions in dimensions $d = 1, 2, 3$ respectively. The case of general d is considered in [Appendix A.3](#).

Theorem 1.107 (Hankel transform) If $u \in L^2(\mathbb{R}^d, \mathbb{C})$ is isotropic, the Fourier transform $\hat{u}(\lambda)$ only depends on $s := \|\lambda\|_2$ and, in the cases $d = 1, 2, 3$, we have

$$\hat{u}(s) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-i\lambda^\top \mathbf{x}} u(\mathbf{x}) d\mathbf{x} = \begin{cases} \sqrt{\frac{2}{\pi}} \int_0^\infty \cos(rs) u(r) dr, & d = 1, \\ \int_0^\infty J_0(rs) u(r) r dr, & d = 2, \\ \sqrt{2\pi} \int_0^\infty \frac{\sin(rs)}{rs} u(r) r^2 dr, & d = 3. \end{cases}$$

Proof To demonstrate the general principle that an isotropic function $u(\mathbf{x})$ has an isotropic Fourier transform $\hat{u}(\lambda)$, let Q be a rotation on $\mathbb{R}^d$. For the change of variable $\mathbf{x} = Q\mathbf{z}$, $u(\mathbf{x}) = u(\mathbf{z})$ and

$$(2\pi)^{d/2} \hat{u}(Q\lambda) = \int_{\mathbb{R}^d} e^{-i(Q\lambda)^\top \mathbf{x}} u(\mathbf{x}) d\mathbf{x} = \int_{\mathbb{R}^d} e^{-i\lambda^\top Q^\top Q \mathbf{z}} u(\mathbf{z}) |\det Q| d\mathbf{z}.$$

As $Q^\top Q = I$, $\hat{u}(Q\lambda) = \hat{u}(\lambda)$. Because $\hat{u}(\lambda)$ is unchanged by rotation, it is a function of $s = \|\lambda\|_2$ and the Fourier transform is isotropic.

We treat the case $d = 3$ and leave $d = 1, 2$ for [Exercise 1.26](#). Consider spherical coordinates $(x_1, x_2, x_3) = r(\cos \psi, \sin \psi \cos \theta, \sin \psi \sin \theta)$ with $-\pi \leq \theta_1 \leq \pi$ and $0 \leq \psi \leq \pi$. Pick $\lambda = [s, 0, 0]^\top$ so that

$$\int_{\mathbb{R}^3} e^{-i\lambda^\top \mathbf{x}} u(\mathbf{x}) d\mathbf{x} = \int_0^\infty \int_{-\pi}^\pi \int_0^\pi e^{-isr \cos \psi} (\sin \psi) d\psi d\theta u(r) r^2 dr.$$

Let $t = -\cos \psi$ so that $dt = \sin \psi d\psi$ and

$$\int_{-\pi}^\pi \int_0^\pi e^{-isr \cos \psi} \sin \psi d\psi d\theta = \int_{-\pi}^\pi \int_{-1}^1 e^{irst} dt d\theta = \frac{2\pi}{rs} 2 \sin(rs).$$

Hence,

$$\int_{\mathbb{R}^3} e^{-i\lambda^\top \mathbf{x}} u(\mathbf{x}) d\mathbf{x} = 4\pi \int_0^\infty \frac{\sin(rs)}{rs} u(r) r^2 dr = (2\pi)^{3/2} \int_0^\infty \frac{J_{1/2}(rs)}{\sqrt{rs}} u(r) r^2 dr. \quad \square$$

1.5 Notes

The Lebesgue integral is usually developed from limits of non-negative functions, for example, Rudin (1987). Our approach follows Zeidler (1995) for Lebesgue integrals and (Dudley, 1999; Yosida, 1995) for Bochner integrals. Neither the Borel σ -algebra nor the product measure (even of two complete measures) is complete in general; our convention is to always use the extension to the complete measure space. [Lemma 1.20](#) is given for example by Dudley (2002, Theorem 4.2.2) for pointwise convergence and this extends to almost sure convergence if the measure space is complete ([Exercise 1.2](#)). The Lebesgue integral is important in probability theory, as we see in [Chapter 4](#), and is developed in many probability text books including (Fristedt and Gray, 1997; Williams, 1991).

Hilbert spaces and the spaces $H'(D)$ are developed in (Renardy and Rogers, 2004; Robinson, 2001; Temam, 1988; Zeidler, 1995). An excellent reference for the Sobolev embedding theorem is Adams and Fournier (2003). Our treatment of non-negative integral operators is more detailed as they are important for the Karhunen–Loève theorem in [Chapter 5](#). (Lax, 2002; Riesz and Sz.-Nagy, 1990) treat Mercer's theorem. To study fractional powers fully, the operator A should be developed as a *sectorial operator*, which says that the spectrum of L lies in a sector of the complex plane; see Pazy (1983). We have given applications to the Laplacian with periodic and Dirichlet conditions and this is suitable for the reaction–diffusion equations studied in later chapters.

The Fourier transform is covered in detail by Sneddon (1995) as a method. The FFT is discussed in Brigham (1988) and mathematical theory is discussed in Koerner (1989). FFT and spectral methods for MATLAB are discussed in Trefethen (2000).

Exercises

- 1.1 For a bounded domain D , prove that $C(\bar{D})$ is complete with respect to the supremum norm $\|u\|_\infty = \sup_{x \in \bar{D}} |u(x)|$. By constructing a suitable Cauchy sequence, show that $C(\bar{D})$ is not complete with respect to the $L^2(D)$ norm.
- 1.2 Let $(X, \mathcal{F}, \mu)$ be a complete measure space, Y be a Banach space, and $f: X \rightarrow Y$ a measurable function. Show that $f = g$ a.s. implies that g is measurable. Give an example to show that the completeness assumption on the measure space is necessary.
- 1.3 Using [Definition 1.21](#), prove that if $u: X \rightarrow Y$ is integrable then

$$\int_X \|u(x)\|_Y d\mu(x) < \infty.$$

- 1.4 a. If $u: \mathbb{R} \rightarrow \mathbb{R}$ is measurable, define simple functions $u_n: \mathbb{R} \rightarrow \mathbb{R}$ such that $u_n(x) \rightarrow u(x)$ for all $x \in \mathbb{R}$.
- b. If $G \in C(\mathbb{R} \times \mathbb{R})$, define simple functions $G_N(x, y)$ of the form

$$\sum_{j,k=1}^N G(x_j, x_k) 1_{F_j}(x) 1_{F_k}(y),$$

for $x_j \in \mathbb{R}$ and $F_j \in \mathcal{B}(\mathbb{R})$, such that $G_N \rightarrow G$ pointwise as $N \rightarrow \infty$.

- 1.5 Give an example of a Lebesgue integrable function $u: \mathbb{R} \rightarrow \mathbb{R}$ that is not Riemann integrable.
- 1.6 By introducing a measure on $\mathbb{Z}$, prove that

$$\lim_{n \rightarrow \infty} \sum_{j \in \mathbb{Z}} u_{n,j} = \sum_{j \in \mathbb{Z}} u_j, \quad (1.48)$$

for sequences $u_{n,j}$ converging to u_j as $n \rightarrow \infty$ and such that $|u_{n,j}| \leq \bar{U}_j$ for a sequence $\bar{U}_j$ with $\sum_{j \in \mathbb{Z}} \bar{U}_j < \infty$.

- 1.7 Evaluate

$$\int_0^b \int_0^a f(x, y) dx dy, \quad \int_0^a \int_0^b f(x, y) dy dx,$$

taking care to observe the order of integration, for

$$f(x, y) = \begin{cases} \frac{xy(y^2 - x^2)}{(x^2 + y^2)^3}, & 0 < x < a, \quad 0 < y < b, \\ 0, & x = 0, \quad y = 0. \end{cases}$$

Why does Fubini's theorem *not* apply?

- 1.8 Using Fubini's theorem, prove that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} dx = 1. \quad (1.49)$$

Hint: first square both sides.

- 1.9 For $f \in C^1([0, 1])$ and $p \geq 1$, prove that

$$\left(\int_0^1 |f(x)| dx \right)^p \leq \int_0^1 |f(x)|^p dx.$$

This is a version of Jensen's inequality (see [Lemma 4.56](#) and §A.4). For $a < b$, show that $L^q(a, b) \subset L^p(a, b)$ for $1 \leq p \leq q$.

- 1.10 On a real Hilbert space H , prove the Cauchy–Schwarz inequality $|\langle u, v \rangle| \leq \|u\| \|v\|$ for $u, v \in H$. Hint: find the minimum of $\|u + \alpha v\|^2$ with respect to $\alpha \in \mathbb{R}$.
- 1.11 Prove that (1.12) is the weak derivative of (1.11) (by showing (1.13) holds) and that $\mathcal{D}u$ does not have a weak derivative in the set of measurable functions from $\mathbb{R} \rightarrow \mathbb{R}$.
- 1.12 a. Let $u(r) = \log|\log r|$ for $r > 0$. Sketch a graph of $u(r)$ and $u'(r)$. Show that $u \in L^2(0, 1/2)$ but $u \notin C([0, 1/2])$ and $u' \notin L^2(0, 1/2)$.
- b. Let $u(x) = \log|\log r|$ for $r = \|x\|_2$ and $D = \{x \in \mathbb{R}^2: \|x\|_2 < 1/2\}$. Show that u has a well-defined weak derivative $\mathcal{D}_i u$ in $L^2(D)$ for $i = 1, 2$. Hence show that $u \in H^1(D)$ and $u \notin C(\bar{D})$.
- 1.13 For $\ell > 0$ and $u \in C^{n+1}([-\ell, \ell])$, show that there exists a polynomial p of degree at most n such that

$$\|u - p\|_{L^2(-\ell, \ell)} \leq \frac{1}{(n+1)!} \|D^{n+1}u\|_{L^2(-\ell, \ell)} \ell^{n+1}.$$

Hint: use Theorem A.1 and the Cauchy–Schwarz inequality.

- 1.14 Prove the Poincaré inequality for the domain $D = (0, \ell)$ by the following methods:
- a. With $K_p = \ell$, by writing $u(x) = \int_0^x u'(s) ds$.
- b. With $K_p = \ell/\pi$, by writing u in the basis $\{\sin(k\pi x/\ell): k \in \mathbb{N}\}$.

Show that $K_p = \ell/\pi$ is the optimal constant.

- 1.15 For Banach spaces X, Y , prove that $L \in \mathcal{L}(X, Y)$ if and only if L is continuous from X to Y . Prove that $\|Lu\|_Y \leq \|L\|_{\mathcal{L}(X, Y)} \|u\|_X$ for $u \in X$.
- 1.16 Let H be a real Hilbert space. Show that

$$2\|u\|^2 + 2\|v\|^2 = \|u + v\|^2 + \|u - v\|^2, \quad u, v \in H.$$

For a closed subspace G of H , let $p_n \in G$ be such that $\|u - p_n\| \rightarrow \inf_{p \in G} \|u - p\|$. Prove that p_n is a Cauchy sequence.

- 1.17 Prove the Riesz representation theorem ([Theorem 1.57](#)). Hint: consider $G = \{u \in H: \ell(u) = 0\}$. Show that $G^\perp$ is one dimensional and apply [Theorem 1.37](#).
- 1.18 Let H be a separable Hilbert space and $B = \{u \in H: \|u\| \leq 1\}$. If H is infinite dimensional, prove the following:
- a. B is not compact.
- b. If $L: H \rightarrow H$ is compact, $L(B)$ is bounded and $L \in \mathcal{L}(H)$.
- c. Let μ be a measure on H such that $\mu(F) = \mu(x + F)$ for $F \in \mathcal{B}(H)$, $x \in H$, and $x + F := \{x + y: y \in F\}$ (we say the measure is *translation invariant*). If $\mu(B)$ is finite, then $\mu(B) = 0$.
- 1.19 Let $L \in \mathcal{L}(H)$ be non-negative. Prove that $\text{Tr } L = \sum_{j=1}^{\infty} \langle L\phi_j, \phi_j \rangle$ is independent of the orthonormal basis $\{\phi_j: j = 1, \dots, \infty\}$.
- 1.20 Calculate the Green's functions for the operators in Examples 1.81 and 1.84.

- 1.21 Prove that the Fourier transform of a convolution $u \star v$, defined by

$$(u \star v)(\mathbf{x}) := \int_{\mathbb{R}^d} u(\mathbf{x} - \mathbf{y})v(\mathbf{y}) d\mathbf{y},$$

for $u, v \in L^1(\mathbb{R}^d)$, is given by

$$\widehat{u \star v}(\lambda) = (2\pi)^{d/2} \hat{u}(\lambda) \hat{v}(\lambda). \quad (1.50)$$

- 1.22 With u_k and U_k defined by (1.35) and (1.39) in the case $(a, b) = (0, \ell)$, show that

$$U_k = \sum_{n \in \mathbb{Z}} u_{k+nJ}, \quad \forall u \in H_{\text{per}}^2(0, \ell).$$

- 1.23 Let $Au := u - u_{xx}$ for $u \in \mathcal{D}(A) := H_{\text{per}}^2(0, 2\pi)$ and denote the associated fractional power norm by $\|\cdot\|_r$. Let J be even and I_J be defined by (1.40). Let P_J denote the orthogonal projection from $L^2(0, 2\pi)$ onto $\text{span}\{e^{2\pi i j x} : j = -J/2 + 1, \dots, J/2\}$.

- a. Using [Exercise 1.22](#), prove that for $r > 1/4$ there exists $K > 0$ such that

$$\|I_J u - P_J u\|_{L^2(0, 2\pi)} \leq \frac{K}{J^{2r}} \|u\|_r, \quad \forall u \in \mathcal{D}(A^r).$$

- b. Prove [Lemma 1.100](#) by showing that, for some $K > 0$,

$$\|u - I_J u\|_{1/2} \leq K \frac{1}{J} \|u\|_1.$$

Hint: use [Lemma 1.94](#) and [Proposition 1.93](#).

- c. Let u_n be given by (1.35) and suppose there exists $K, p > 0$ such that $|u_n| \leq K|n|^{-(1+p)}$ for all $n \in \mathbb{Z}$. Show that

$$\left| \|u\|_{r/2} - \|I_J u\|_{r/2} \right| = \mathcal{O}(J^{-(1+p)}), \quad 0 < 2r \leq p.$$

Now prove that $\|u_1\|_{H^1(0, 2\pi)} - \|I_J u_1\|_{H^1(0, 2\pi)} = \mathcal{O}(J^{-3} \log J)$ for the function $u_1(x)$ shown in [Figure 1.6](#).

- 1.24 a. Prove that the trapezium rule approximation $T_J(u^2)$ to $\int_0^1 u(x)^2 dx$ can be written as $T_J(u^2) = \|I_J u\|_{L^2(0, 1)}^2$ for any $u \in C(0, 1)$ such that $u(0) = u(1)$.
 b. By using [Theorem A.6](#), explain the higher rates of convergence shown in [Figure 1.6\(a\)](#) for the $L^2(0, 1)$ norm.
- 1.25 Consider $u \in L^2(0, 2\pi)$. Suppose that the Fourier coefficient $u_k = \mathcal{O}(|k|^{-(2r+1+\epsilon)/2})$, for some $r, \epsilon > 0$. Show that $u \in H^r(0, 2\pi)$.
- 1.26 Derive the formula for the Hankel transform given in [Theorem 1.107](#) for $d = 1, 2$.
- 1.27 The discrete cosine transform (DCT-1) $\mathbf{y} \in \mathbb{R}^N$ of a vector $\mathbf{x} \in \mathbb{R}^N$ is defined by

$$y_k := \frac{1}{2}(x_1 + (-1)^{k-1} x_N) + \sum_{j=2}^{N-1} x_j \cos\left(\frac{\pi(k-1)(j-1)}{N-1}\right),$$

for $k = 1, \dots, N$. Show how to evaluate $\mathbf{y}$ by using the FFT on a vector of length $2N - 2$, which you need to specify in terms of x_j .

The discrete sine transform (DST-1) $\mathbf{y} \in \mathbb{R}^N$ of $\mathbf{x} \in \mathbb{R}^N$ is defined by

$$y_k := \sum_{j=1}^N x_j \sin\left(\frac{\pi j k}{N+1}\right), \quad k = 1, \dots, N.$$